Technical Safety Concept

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1 Technical Safety Requirements

1.1 Ambarella Chip SW Safety Requirements

1.1.1 BIST TSRs

MVC-99658 - LBIST Vector Transfer to IST SRAM (SM_HW_LBIST)

The SoC boot software shall copy the LBIST vector data from the off chip non volatile memory (NAND flash) into the **In System Test (IST) SRAM** after power on reset.

Note: Vector data is provided in Ambarella bootstrap reference code.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Put a breakpoint in the LBIST routine to make sure the routine is called at startup.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111228 - LBIST Vector Transfer to IST SRAM (SM_HW_LBIST) is satisfied by : MVC-111609 - Enabling the LBIST mechanism

MVC-99659 - LBIST Controller Activation (SM_HW_LBIST)

The SoC boot software shall write to the **IST CONTROL register** to kick off the LBIST operation for each LBIST controller in the SoC.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review the boot source code to confirm that sequence of writes to all IST CONTROL registers is present.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111229 - LBIST Controller Activation (SM_HW_LBIST)

MVC-99663 - PO_MBIST Vector Transfer (SM_HW_PO_MBIST)

The SoC boot software shall copy the Power On Memory BIST (PO_MBIST) library binary into the **IST SRAM** after every power on reset.

Note: Vector data is provided in Ambarella bootstrap reference code.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Put a breakpoint in MBIST routine to make sure the routine is called at startup.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111608 - Enabling the MBIST mechanism

MVC-99662 - PO_MBIST Vector Activation (SM_HW_PO_MBIST)

The SoC boot software shall write to the IST CONTROL registers to initiate MBIST execution for the ECC protected SRAMs in the VMEM and SMEM clusters.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review the boot source code to confirm that writes to all IST CONTROL registers is present.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs

MVC-99665 - Reading LBIST and MBIST Completion Status (SM_HW_LBIST, SM_HW_PO_MBIST)

The SoC boot software shall read the IST STATUS register to determine the completion and result of the LBIST and MBIST operations.

If the IST STATUS register indicates a **PASS** status for all LBIST and MBIST tests, the SoC SW continue with normal operation.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Verify that when the status register indicates no failures, the SoC SW proceeds to normal operation.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111232 - Reading LBIST and MBIST Completion Status (SM_HW_LBIST, SM_HW_PO_MBIST) is satisfied by : MVC-111233 - Reading LBIST and MBIST Completion Status (SM_HW_LBIST, SM_HW_PO_MBIST)

MVC-99664 - Reaction to LBIST and MBIST Failure (SM_HW_LBIST, SM_HW_PO_MBIST)

If the IST STATUS register indicates a **FAIL** status for any LBIST controller OR a memory fault for any SRAM under test, the SoC SW shall assert a safety failure and go to safe state within 200ms.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Use the debugger to simulate a failure indicated by IST STATUS register then resume execution. - Verify that the SoC SW asserts a safety error and goes to safe state.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111234 - Reaction to LBIST and MBIST Failure (SM_HW_LBIST, SM_HW_PO_MBIST)

MVC-99666 - Reading CEHU Initial Safety Error Status (SM_SW_CHK_SAFETY_ERR)

After every power on reset or global reset, and before clearing any error flags, the SoC software shall read the **CEHU registers** and **ECRU registers** to capture the initial safety error state of all components.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review the reset handling source code to confirm that the first operations after reset are reads of the CEHU and ECRU registers.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111237 - Reading CEHU Initial Safety Error Status (SM_SW_CHK_SAFETY_ERR)

MVC-99667 - Verification of Expected Post Reset Error Vector Values (SM_SW_CHK_SAFETY_ERR)

The SoC software shall compare the values read from the CEHU **error_bit_vector** registers and ECRU registers against the expected post reset pattern.

Note: Post-reset pattern is defined in the Safety Manual section 5.2.2

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Boot the system normally - After comparison is done successfully, verify that the system proceeds to normal operation - Inject a corrupted value to a selected CEHU or ECRU register - Run the software again - Verify that CEHU error is reported.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111236 - Verification of Expected Post Reset Error Vector Values(SM_SW_CHK_SAFETY_ERR)

MVC-99668 - Reaction to Inconsistent Initial Safety Error State (SM_SW_CHK_SAFETY_ERR)

If the values read from the CEHU or ECRU registers are inconsistent with the expected post reset state, the system shall go to safe state within 200ms (Normal operation is not possible)

Note: For the expected register values, refer to Ambarella Safety Manual, section 5.2.2 "Software Check Initial Safety Error State"

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI

Verification Criteria	- Use the debugger to simulate a failure indicated to a selected CEHU or ECRU register - Run the software again - Verify that the system goes to safe state within time constraint
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111239 - Reaction to Inconsistent Initial Safety Error State (SM_SW_CHK_SAFETY_ERR)

MVC-99669 - Clearing of Initial Safety Error Status (SM_SW_CHK_SAFETY_ERR)

If the initial safety error state matches the expected pattern, the SoC software shall clear the CEHU and ECRU error status registers to prepare for runtime error monitoring.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Boot the system normally - After comparison is done successfully, Read the current values of the CEHU/ECRU registers - After the clear, read the registers again, verify the CEHU/ECRU registers are now cleared.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99727 - BIST TSRs is related to : ~~MVC-105433~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - BIST TSRs is satisfied by : MVC-111238 - Clearing of Initial Safety Error Status (SM_SW_CHK_SAFETY_ERR)

1.1.2 Clock TSRs

MVC-99671 - initialization of the hardware clock monitors (SM_HW_CKMON)

The SoC Software shall initialize hardware clock monitors by:

1. verifying clock monitors error flags to CEHU after power-on reset,
2. clearing CEHU clock monitor error bits,
3. programming PLLs and clock dividers
4. setting each clock monitor's upper_bound and low_bound thresholds via configuration registers
5. programming CEHU to receive errors, and
6. enabling each clock monitor.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - wait util system completed boot-up - observe: CEHU monitor errors are cleared - observe: PLLs and clock deviders are programmed - observe: clock monitor's upper_bound and low_bound thresholds are set - observe: CEHU is configured to receive errors - observe: enable input of clock monitors is set
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99726 - Clock TSRs is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs is satisfied by : MVC-111243 - Initialization of the hardware clock monitors (SM_HW_CKMON)

MVC-99670 - Stuck clocks detection (SM_HW_CKMON)

The hardware clock monitor shall detect and indicate errors (by signaling to CEHU error pin(s) or signalling to SW interrupt - depending on configuration of error routing) for out-of-bounds or stuck clocks within 99 ms maximum.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW

Verification Criteria	- Run system with all clocks at nominal frequencies. - Read the CEHU error register; verify no clock monitor errors - Decrease the frequency of the monitored clock until it falls below the threshold - Start a timer immediately - Measure the time from threshold crossing to error assertion. - Verify this time is <= 99ms - Run system (again) with all clocks at nominal frequencies. - Read the CEHU error register; verify no clock monitor errors - Increase the frequency of the monitored clock until it reached the level above the threshold - Start a timer immediately - Measure the time from threshold crossing to error assertion. - Verify this time is <= 99ms
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99726 - Clock TSRs is satisfied by : MVC-101036 - Stuck clocks detection is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs

MVC-99673 - Reaction of the stuck clocks (SM_HW_CKMON)

If stuck is detected, the SoC SW shall transition into SafeState within 200ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run system with all clocks at nominal frequencies. - Read the CEHU error register; verify no clock monitor errors - Decrease the frequency of the monitored clock until it falls below the threshold - Start a timer immediately - Measure the time from threshold crossing to error assertion. - Verify the time constraint
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99726 - Clock TSRs is satisfied by : MVC-101038 - Reaction of the stuck clocks is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs is satisfied by : MVC-111241 - Reaction of the stuck clocks (SM_HW_CKMON) is satisfied by : MVC-111242 - Reaction of the stuck clocks (SM_HW_CKMON)

MVC-99672 - Nodes to be enabled (SM_HW_CKMON)

The SoC software shall enable clock monitoring for the following nodes:

- PLL outputs
- Clock divider outputs
- Cluster level clock gating cell output
- Right before Cortex-R52 CLKIN input
- Clock to main logic of lockstep / duplicate function

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review the clock monitor initialization source code to confirm that writes to the enable registers include all the required bits.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... has parent : MVC-99726 - Clock TSRs is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs is satisfied by : MVC-111245 - Nodes to be enabled (SM_HW_CKMON)

MVC-99675 - Steps during startup test (SM_SW_CKMON_TEST)

During startup test, SoC software shall test each hardware clock monitor by:

1. enabling clock source,
2. setting clock monitor upper_bound to zero and lower_bound to max via configuration registers,
3. programming CEHU to receive error,
4. Enable each clock monitor
5. verifying CEHU receives the error within 99ms
6. If SoC SW receives error signal (within 99ms), the SoC SW shall clear the relevant (CEHU error source)bits.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	for each clock monitor: - start the system - pause the system - make sure clock source is anabled - set clock monitor upper_bound to zero and lower_bound to max via configuration registers - resume start up - observe CEHU receives an error withn 99ms - observe the relevant (CEHU error source)bits has been cleared
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99726 - Clock TSRs is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs is satisfied by : MVC-111244 - Steps during startup test (SM_SW_CKMON_TEST)

MVC-99674 - Reaction on the missing receiving error in CEHU (SM_SW_CKMON_TEST)

If (as a result of the sequence depicted in MVC-99675) error from clock monitor not received in CEHU, SoC SW shall transition the System into SafeState within 200ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Before the test runs, disable the clock monitor's ability to report errors - Resume execution and allow the test to run - Verify that no error appears in CEHU - observe SoC SW transitions the System into SafeState wihin time constraints
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99726 - Clock TSRs is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs is satisfied by : MVC-111246 - Reaction on the missing receiving error in CEHU (SM_SW_CKMON_TEST) is satisfied by : MVC-111247 - Reaction on the missing receiving error in CEHU (SM_SW_CKMON_TEST)

MVC-99677 - Provide 24 MHz clock with 50 ppm tolerance.

The system shall provide a 24 MHz reference clock to the SoC CLKIN pin with ±50 ppm tolerance.

Note: ref. to [AR-05] in "Ambarella CV2FS CV22FS CV2FSA HW Safety Manual" (Ambarella CV2FS CV22FS HW Safety Manual (rev. 1522576), HW-SM-002)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start the system - measure the frequency (in room temperature) of the clock device - verify that the nominal frequency and tolerances are met - verify by the manufacturer the temperature and long life stability of the clock device P.S. for the measurement use industry standard procedures if applicable.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99726 - Clock TSRs is satisfied by : MVC-101035 - Provide 24 MHz clock with 50 ppm tolerance is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs is verified by : MVC-111813 - CLK_Ref_CLKIN_001

MVC-99676 - The 32.768 kHz RTC clock usage as reference clock

The system shall provide a 32.768 kHz RTC clock to the SoC with ±100 ppm tolerance to cross check the accuracy of the 24 MHz reference clock.

Note: ref. to [AR-06] in "Ambarella CV2FS CV22FS CV2FSA HW Safety Manual" (Ambarella CV2FS CV22FS HW Safety Manual (rev. 1522576), HW-SM-002)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start the system - measure the frequency (in room temperature) of the clock device - verify that the nominal frequency and tolerances are met - verify by the manufacturer the temperature and long life stability of the clock device P.S. for the measurement use industry standard procedures if applicable.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99726 - Clock TSRs is satisfied by : MVC-101034 - Provide 32.768 kHz clock with 100 ppm tolerance is related to : ~~MVC-105434~~ - [TechSaReqs] Ambarella Chip SW Safety Requirements - Clock TSRs is verified by : MVC-111814 - CLK_RTC_CLK_001

1.1.3 Power TSRs

MVC-108681 - System reaction on power monitor reported error (SM_HW_PSMON)

Before SM_HW_PSMON configuration, the SoC SW shall configure the CEHU SAFETY_MODE register, such that if a power monitor error is reported to CEHU, the CEHU will propagate the error to the SoC safety pin.

For details, refer to Ambarella Safety Manual section "4.3 Error Handling"

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Verified in another requirement MVC-99681 - Additionally, read the CEHU configuration SAFETY_MODE before configuration SM_HW_PSMON is executed, to verify that power monitor error is routed to safety pin
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99728 - Power TSRs is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs is satisfied by : MVC-111249 - System reaction on power monitor reported error (SM_HW_PSMON)

MVC-99679 - Initialization the Hardware Power Supply Monitor (SM_HW_PSMON)

The SoC software shall initialize the SoC internal Hardware Power Supply Monitor by the following sequence of steps:
(1) Setup of the the related power monitor register (power down register, disable control register)
(2) Selecting the input voltage sensing levels for each power rail
(3) Programming the under voltage and over voltage comparison reference levels
(4) Clearing old result by toggling the reset control register.
(5) Enabling the sticky error function

The following power rails used by SoC shall be monitored:

- PMIC buck 1 (SoC VDDI)
Nominal: 0.75V, Minimum: 0.73V, Maximum: 0.77V

- PMIC buck 2 (SoC VDDI)
Nominal: 0.75V, Minimum: 0.73V, Maximum: 0.77V

- PMIC buck 3 (SoC DDR_DVDD_VDDQ, SoC DDR_DVDD_IO)
Nominal: 1.1V, Minimum: 1.06V, Maximum: 1.17V

- PMIC buck 4 (SoC AVDD)
Nominal: 0.8V, Minimum: 0.78V, Maximum: 0.825V

- PMIC buck 5 (SoC AVDD18)
Nominal: 1.8V, Minimum: 1.7V, Maximum: 1.9V

For details on configuration steps, refer to:
- Ambarella Safety Manual section "5.1.1 Power Monitor"
- Ambarella Jira https://jiraap.ambarella.com/browse/MEKRALANGPOC-373

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B

Allocation	HSI
Verification Criteria	- After each configuration step, read the configured registers to verify that they are written with the correct values and in the specified order. - Step 5 is verified in requirement MVC-99681
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... relates to : MVC-112498 - PMIC Bucks routing/naming changed - update requirements on System Level has parent : MVC-99728 - Power TSRs is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs is satisfied by : MVC-111248 - Initialization the Hardware Power Supply Monitor (SM_HW_PSMON)

MVC-99678 - Monitoring the configured power supply rails (SM_HW_PSMON)

The SoC internal Hardware Power Supply Monitor shall continuously monitor the configured power supply rails.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- At nominal voltage, read the CEHU status register, verify no PSMON errors - Decrease the voltage of a monitored rail until it crosses below UV threshold. - Observe the CEHU status register - verify that safety error signal is asserted - Increase the voltage of a monitored rail until it crosses above OV threshold. - Observe the CEHU status register - verify that safety error signal is asserted
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99728 - Power TSRs is satisfied by : MVC-101044 - Monitoring the configured power supply rails. is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs

MVC-99681 - Reaction on under voltage, over voltage conditions (SM_HW_PSMON)

If under voltage or over voltage conditions are detected, the SoC internal HW power supply monitor shall assert power monitor error signal pm_errorb15 to the Central Error Handling Unit (CEHU) and asset low the safety pin within 100us

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- At nominal voltage, read the CEHU status register, verify no PSMON errors - Verify the SoC safety pin is asserted high - Decrease the voltage of a monitored rail until it crosses below UV threshold. - Start a timer immediately - Observe the CEHU status register - Verify that power monitor error signal is asserted - Measure the time from when the voltage crosses the UV threshold to when safety error signal is asserted - Verify the SoC safety pin is asserted low
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99728 - Power TSRs is satisfied by : MVC-101046 - Reaction on under voltage, over voltage, or voltage drift conditions is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs

MVC-99680 - Perform a latent fault test (SM_SW_PSMON_TEST)

The SoC software shall perform a latent fault test for each SoC powered up internal power monitor at the start of each power cycle.

Refer to Ambarella SoC Safety Manual section 5.2.1 for details.

Status	Draft
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Place the selected power monitor into test mode, - Force an out of range condition by programming illegal under voltage/over voltage thresholds, - Trigger a measurement cycle, - Read the observation status register and the CEHU safety error status register, - Verify that the expected error is correctly reported to the CEHU, - Verify test mode is exited (errors are cleared, operational voltage ranges restored)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99728 - Power TSRs is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs is satisfied by : ~~MVC-111251~~ - Perform a latent fault test (SM_SW_PSMON_TEST)

MVC-99683 - Reaction on the missing receiving error in CEHU (SM_SW_PSMON_TEST)

If an error from any SoC powered up internal power supply monitor is not received by CEHU immediately after the test is performed, the SoC SW shall assert a safety error and shall go to safe state within 200ms (FHTI).

Refer to Ambarella SoC Safety Manual section 5.2.1 for details.

Status	Draft
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Before the test runs, disable the power monitor's error input in CEHU - Resume execution and allow the test to run - Verify that no error appears but software asserts safety error
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99728 - Power TSRs is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs is satisfied by : MVC-111250 - Reaction on the missing receiving error in CEHU (SM_SW_PSMON_TEST)

MVC-99682 - Implementation of an external power monitor for AVDD18 power supply

The hardware shall implement an external power monitor for AVDD18 power supply to monitor over-voltage and under-voltage.

Definitions:
Nominal voltage = 1.8V
Under-voltage threshold = 1.7V
Over-voltage threshold = 1.9V

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Verified in another requirement MVC-99684 - Additionally, schematics shall be reviewed to check the existence of AVDD18 power monitoring circuitry.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99728 - Power TSRs is satisfied by : MVC-101048 - Implementation of an external power monitor for AVDD18 power supply is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs

MVC-99684 - Detection of under voltage or over voltage conditions

If under voltage or over voltage conditions are detected, the external AVDD18 power monitor circuit shall power off all power rails used by SoC as listed below, within 50ms.
- PMIC buck 1 (0.75V)
- PMIC buck 2 (0.75V)
- PMIC buck 3 (1.1V)
- PMIC buck 4 (0.8V)
- PMIC buck 5 (1.8V)

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- With AVDD18 at nominal, verify that all SoC power rails are at their proper voltages and the system is running - Decrease AVDD18 below the under-voltage threshold. - Verify that the SoC power rails reach 0V - Verify the time constraints - Repeat the steps above for over-voltage test
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... relates to : MVC-112498 - PMIC Bucks routing/naming changed - update requirements on System Level has parent : MVC-99728 - Power TSRs is satisfied by : MVC-101047 - Detection of under voltage, over voltage, or drift conditions is related to : MVC-105435 - [TechSaReqs] Ambarella Chip SW Safety Requirements - Power TSRs

1.1.4 Other SoC Safety Mechanisms

MVC-115351 - Test Frame (SM_HW_TESTFRAME)

The system shall configure the execution period of the DSP Test Frames.
Note: This period must be defined such that the sum of the Test Frame execution time and the software overhead remains below the Fault Tolerant Time Interval (FTTI) of the relevant system-level Safety Goal.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115350 - Test Frame Error Detection (SM_HW_TESTFRAME)

The System shall implement and register a high-priority callback function (Error Handler) with the SDK for capturing asynchronous error notifications generated by a Test Frame failure in the DSP engine.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115352 - Test Frame Error Reaction (SM_HW_TESTFRAME)

Upon detection of Test Frame failure in the DSP engine, the system shall go to safe state within 200ms.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115354 - VIN ORC SW Readable Status Register (SM_HW_VIN_STATUS)

The system shall monitor the Video Input (VIN) interface to detect communication errors reported by the hardware status registers.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115353 - VIN ORC SW Readable Status Register Error Detection (SM_HW_VIN_STATUS)

the system shall define which error types or frequencies constitute a failure based on analyzing the impact of VIN errors (e.g., a single CRC error vs. a persistent FIFO overflow) on the system-level Safety Goal.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115355 - VIN ORC SW Readable Status Register Error Reaction (SM_HW_VIN_STATUS)

Upon detection of a defined failure condition, the system shall transition to a Safe State within 200ms

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115342 - Error Counter and Threshold to Detect Permanent Fault in Pixel Data SRAM (SM_HW_ERR_COUNTER_THR)

The system shall implement an address and data parity checks against a programmed threshold for the SRAMs in the IDSP function.

Note: the aim is to detect permanent faults in the IDSP line buffer SRAMs.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	Configure the SRAM parity error threshold with a known value, inject address and data parity errors, and verify that: (1) the error counter increments accordingly (2) the CEHU indicates the safety error only when the accumulated parity error count exceeds the programmed threshold.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115344 - Error Counter and Threshold reaction upon detection of Permanent Fault in Pixel Data SRAM (SM_HW_ERR_COUNTER_THR)

Upon detection of the parity error, then the system shall transition to safe state within 100ms

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	Verify that the system transitions to the Safe State within 100 ms of detecting the parity error.
Verification Method	Testable, checked by Test

Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115357 - Fault Injection Logic (SM_HW_FAULT_INJ)

The system shall provide a fault injection mechanism for each safety-relevant error signal collected by the AMBA ECRU, except for error signals originating from Cortex-A53 and Cortex-R52 that do not support fault injection.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115356 - CPU Error Monitoring Unit (SM_HW_CEMU)

The system shall implement and integrate the software for detecting any wrong address and L2 error signals from Cortex-A53.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115359 - Fault Injection for DCLS in Cortex-R52 (SM_HW_FAULT_INJ_CR52)

The system shall detect the latent fault in Cortex R52 by the DCLS design.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

MVC-115358 - SRAM SECDED for Address and Data (SM_SRAM_SECDED_ADDR_DATA)

The System shall ensure that CEHU error reports from SECDED-protected SRAMs are handled by the safety application on the Cortex-R52.

Note: A DBE event shall be treated as a critical, non-recoverable hardware fault requiring immediate transition to a Safe State.

Status	Draft
Assignee(s)	
ASIL	
Allocation	
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-115343 - Other SoC Safety Mechanisms

1.2 Safety Goal Requirements

1.2.1 SG_1 TSRs

MVC-99783 - Check of image frame delay

The SoC software shall check that the total image delay of the SOW2 system from start of every image frame acquisition (SoC internal event from 'RAW Data' Buffer) to start of output of the same image frame (Event from 'Compressed Video'-Buffer) does not exceed 50ms.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run the system under normal conditions. - Capture both the acquisition start event and the output start event - Measure the time difference for ## consecutive frames. - Verify that the maximum measured delay are <= 50 ms.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99782 - SG_1 TSRs is related to : MVC-111535 - [TechSaReqs] SafetyGoal 1 Requirements

MVC-113312 - VCYNC usage in SW

The SoC SW shall utilize the VSYNC signal at GPIO (input) pin V16 of R52 safety core.

Note: the VSYNC signal will be used for start-of-frame determination (timestamp).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	review a relevant part of coude for VSYNC reception/usage routine.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99782 - SG_1 TSRs

MVC-115162 - VSYNC usage in HW

The SoC HW shall route the VSYNC signal at pin V16 of R52 safety core.

Note: the VSYNC signal will be used for start-of-frame determination (timestamp).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start the system, - observe on V16 pin VSYNC signal being transmitted
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99782 - SG_1 TSRs

MVC-113311 - Read Frame Number.

The SoC SW shall retrieve the current frame number from the SOW1 imager upon the arrival of every VSYNC Signal.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	-Trigger the arrival of a VSYNC signal in the system. -Verify that the SoC software executes a read operation of the frame number from the SOW1 imager upon receiving the VSYNC signal. -Verify that the read frame number operation is successfully retrieved and stored or processed by the safety software for sequence verification.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99782 - SG_1 TSRs

MVC-99784 - Confirmation of image frame delay fault

The SoC software shall confirm an image delay fault in case the delay is above the threshold for a duration of 100ms (FDTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run the system under normal conditions. - Inject delay > 50ms for ## consecutive frames - Monitor fault confirmation flag - Measure time from first delayed frame to fault confirmation - Compare against required FDTI (100ms) Note: FDTI will be stretched over consecutive frames duration, as such depends on frame rates.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99782 - SG_1 TSRs is related to : MVC-111535 - [TechSaReqs] SafetyGoal 1 Requirements

MVC-99785 - Reaction of image frame delay fault

Once an image delay fault is confirmed, the system shall complete the transition to the safe state within 200ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz

ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run the system under normal conditions. - Force delay > 50ms for ## consecutive frames such that FDTI is exceeded - Start timer immediately after fault is confirmed - Monitor Ethernet PHY activity- Stop timer when PHY becomes inactive (no data transmission) - Compare the time from the start till PHY become inactive <= 200ms
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99782 - SG_1 TSRs is related to : MVC-111535 - [TechSaReqs] SafetyGoal 1 Requirements

1.2.2 SG_2 TSRs

MVC-99791 - Configuration of timeout mechanism for input frame (SM_SW_WDT_VIN)

The SoC software shall configure a watchdog timer to verify that a new frame is received from SOW1 according to the configured frame rate, with a timeout threshold calculated as 1/(expected input frame rate * 0.8) (incorporating a 20% margin).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Set the expected input frame rate coming from SOW1. -Verify that the software correctly calculates the watchdog timeout value based on this frame rate (including the 20% safety margin). -Verify that the watchdog timer is successfully programmed with this value.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99804 - Enabling the watchdog timer for first input frame (SM_SW_WDT_VIN)

The SoC software shall enable the watchdog timer after VIN and MIPI-DPHY are configured to receive image data

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Start the system and let it finish configuring the VIN and MIPI-DPHY interfaces. -Verify that the software enables the watchdog only after these interfaces are ready to receive image data.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99806 - Frame reception timeout reset (SM_SW_WDT_VIN).

The SoC software shall reset the frame reception timer each time a new frame is received from SOW1.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Start receiving image frames from SOW1. -Verify that whenever a new frame arrives, the watchdog timer is immediately reset to start counting from zero again.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99805 - Detection of timeout by reading the VIN Watchdog Status Register (SM_SW_WDT_VIN).

The SoC software shall detect a timeout by reading the VIN Watchdog Status Register.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Intentionally stop or delay incoming image frames to simulate a timeout condition. -Verify that the software periodically reads the VIN Watchdog Status Register. -Verify that the software successfully detects the timeout flag from the register.

Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99807 - Reaction of frame reception timeout (SM_SW_WDT_VIN).

If the frame reception timeout fault is detected for a duration of 100ms (FDTI), the system shall confirm the fault AND complete the transition to the safe state within 200ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Keep the frame reception timeout condition active. -Verify that the system waits for the specified duration to confirm the fault (FDTI). -Verify that the system completes the transition to the safe state within the required time window (FRTI).
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99809 - Configuration of timeout mechanism for output frame

The SoC software shall configure a timer-based monitoring mechanism to verify that a new output frame is packaged to be transmitted (Event from 'Compressed Video'-Buffer) every 1/(expected output frame rate * 0.8). The check shall take place on each output stream separately in case of dual output streams.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Set the expected output frame rate for each output stream. -Verify that the software calculates the timer-based monitoring threshold using the formula 1/(expected output frame rate * 0.8) to incorporate the safety margin. -Verify that the timer-based monitoring mechanism is configured separately for each stream in case of dual output streams.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99808 - Frame transmission timeout reset

The SoC software shall reset the output frame timer each time the image frame is packaged for Ethernet transmission.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Trigger the packaging of an image frame for Ethernet transmission from the 'Compressed Video' buffer. -Verify that the software immediately resets the output frame timer each time a new frame is successfully packaged.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99810 - Reaction of frame transmission timeout

If the frame transmission timeout fault is detected for a duration of 100ms (FDTI), the system shall confirm the fault AND complete the transition to the safe state within 200ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	-Stop the frame from the 'Compressed Video' buffer to provoke a transmission timeout. -Verify that the system waits for the specified duration of 100ms to confirm the fault (FDTI). -Verify that the system completes the transition to the safe state within an additional 200ms (FRTI).
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99811 - Compare the current received image counter with the previous received image counter

The SoC software shall compare the current received image counter with the previous received image counter and verify that the increment is exactly 1.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Start streaming and inject a frame with counter value = previous + 2 - Verify software detects increment ≠ 1 and sets fault flag. - Inject a frame with same counter value as previous - Verify software detects increment = 0 and sets fault flag - Inject a frame with counter value less than previous - Verify software detects decrement and sets fault flag.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-95914 - The system shall check the strict monotonicity (with increment exactly: 1) of th... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99812 - Reaction on received image counter increment

If the received image counter increment is not 1, and this fault is detected for a duration of 100ms (FDTI), the system shall confirm the fault AND complete the transition to the safe state within 200ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Inject continuous counter errors for more than 100ms. - Start timer at first frame with incorrect counter increment. - Verify flag is set at 100ms - Measure time from fault confirmation to PHY inactive. - Verify this time ≤200ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-95914 - The system shall check the strict monotonicity (with increment exactly: 1) of th... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99875 - Compare the current outgoing image counter with the previous outgoing image counter

The SoC software shall compare the current outgoing image counter with the previous outgoing image counter and verify that the increment is exactly 1. The check shall take place on each output stream separately in case of dual output streams.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Start streaming and inject a frame with counter value = previous + 2 - Verify software detects increment ≠ 1 and sets fault flag. - Inject a frame with same counter value as previous - Verify software detects increment = 0 and sets fault flag - Inject a frame with counter value less than previous - Verify software detects decrement and sets fault flag.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99874 - Reaction on outgoing image counter increment

If the outgoing image counter increment is not 1, and this fault is detected for a duration of 100ms (FDTI), the system shall confirm the fault AND complete the transition to the safe state within 200ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Inject continuous counter errors for more than 100ms. - Start timer at first frame with incorrect counter increment. - Verify flag is set at 100ms - Measure time from fault confirmation to PHY inactive. - Verify this time ≤200ms.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99873 - Compare the current output image counter with the previous output image counter

~~The SoC software shall compare the current output image counter with the previous output image counter and verify that the increment is exactly 1~~.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Start streaming and inject a frame with counter value = previous + 2 - Verify software detects increment ≠ 1 and sets fault flag. - Inject a frame with same counter value as previous - Verify software detects increment = 0 and sets fault flag - Inject a frame with counter value less than previous - Verify software detects decrement and sets fault flag.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

MVC-99872 - Reaction by assertion a safety error

~~If the output image counter increment is not 1, and this fault is confirmed after a duration of 100ms (FDTI), the system shall complete the transition to the safe state within 200ms (FRTI)~~.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Inject continuous counter errors for more than 100ms. - Start timer at first frame with incorrect counter increment. - Verify flag is set at 100ms - Measure time from fault confirmation to PHY inactive. - Verify this time ≤200ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99793 - SG_2 TSRs is related to : MVC-111536 - [TechSaReqs] SafetyGoal 2 Requirements

1.2.3 SG_3 TSRs

MVC-100070 - Configure IDSP control channel

The SoC software shall configure IDSP control channel (additional processing channel) for processing a static stripe set test pattern.

Note: The Static Stripe Set Test Pattern is a fixed test image containing color stripes, gray ramps, and fine black-and-white lines. The system uses this static pattern to check that the image processor keeps colors accurate, gradients smooth, and fine details crisp without blurring or distortion.



Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	- After system initialization, verify that the channel is enabled and correctly configured. - Read the memory location pointed to by the input buffer address. - Verify that the pattern data is present
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100069 - Load a predefined static test pattern stripe set into a reserved DRAM buffer

The SoC software shall load a predefined static test pattern stripe set into a reserved DRAM buffer during system initialization.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	Inspect that the predefined static test pattern stripe set is copied into the reserved DRAM buffer

Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100072 - IDSP parameter set

The SoC software shall use the same IDSP parameter set for control channel and the respective video channel.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	Inspect that the same configuration data structure is used to program both channels.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100071 - Switch dewarping off

The SoC software may switch dewarping off for the control channel.

Status	Rejected
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	Inspect that dewarping enable/disable is implemented separately per channel.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100074 - Setting the control channel's ring buffer

The SoC software shall set the control channel's ring buffer to reference the DRAM location containing the static test pattern stripes

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	Inspect that the ring buffer pointer is set to the correct test pattern buffer address during initialization.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100073 - Control channel to execution

The SoC software shall schedule each control channel to execute periodically every 3rd frame, after the respective main video channel has completed processing a frame.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	- Run system and measure the interval between consecutive control channel executions. - With 30 fps (33.3ms per frame), verify that the period should be approximately 100ms (3 frames). - Count the number of main channel completions between consecutive control channel executions. - Verify this count equals exactly 3.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100076 - Output of the control channel

The SoC software shall capture the output of a scheduled control channel from the 'Preview Data' buffer.

Status	Predefined

Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	- Run system and verify that software performs a read of the Preview Data buffer after control channel completion. - Dump the captured data from the Preview Data buffer. - Compare and confirm against the expected processed test pattern.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100075 - Verify the monotonicity

The SoC software shall verify that the monotonic property of each color ramp and gray ramp in the processed test pattern stripes is preserved.

Note: Monotonicity verification ensures that each color ramp (Red, Green, Blue) and the black-and-white/gray ramp progress smoothly and consistently across the stripe (either strictly increasing or decreasing in intensity) without unexpected sudden inversions, dips, or color spikes (such as a monotonicity defect). This confirms that the IDSP is processing color gradients and brightness levels accurately without distortion.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	SW
Verification Criteria	- Run system, after capture, inject one pixel in a ramp to a lower value than the previous pixel. - Verify software detects monotonicity violation and sets fault flag.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100066 - Verify the color of each patch

The SoC software shall verify that the color of each patch in the processed test pattern stripes remains within expected tolerances.

Note: This check ensures that the measured color and brightness values of each individual color patch (in the RGB and Gray / black-and-white stripes) do not drift or deviate beyond the pre-defined acceptable limits (tolerances) after being processed by the system. It guarantees that color fidelity and accuracy are strictly maintained during live processing.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	SW
Verification Criteria	- Run system, after capture, inject one color patch to have RGB values outside tolerance. - Verify software detects color deviation and sets fault flag.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100068 - Verify the high frequency images

The SoC software shall verify that the high frequency stripe in the processed test pattern is not excessively attenuated.

Note: This check ensures that the fine lines and detailed textures (high-frequency stripes) in the image do not suffer from excessive blurring or signal loss (attenuation) during processing. It guarantees that sharp edges and fine details remain crisp and well-defined without being washed out by the pipeline filters.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	SW
Verification Criteria	- Run system , after capture, Apply low pass filtering to reduce high frequency content significantly (attenuation > threshold). - Verify software detects excessive attenuation and sets fault flag.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100067 - Reaction on the verification checks

If any of the verification checks (monotonicity, color, high frequency) fail, and the fault is confirmed after 150ms (FDTI), the system shall complete the transition to the safe state within 50ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL A
Allocation	HSI
Verification Criteria	- Run system, Inject (non monotonicity, color patch deviation, excessive attenuation in high frequency for 200ms. - Start timer at first failed frame. - Verify fault confirmation flag sets at 150ms. - Measure time from confirmation to PHY inactive. - Verify this time is <= 50ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-100755 - The system shall protect the sent configuration data to camera over I2C using CR...

The system shall protect the sent configuration data to camera over I2C using CRC and message counter.

Note:

The configuration includes the dynamic parameters for AE and AWB.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	Verify that I2C communication between SoC and camera contains CRC and message counter.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111537 - [TechSaReqs] SafetyGoal 3 Requirements

MVC-99786 - Enabling the MIPI CSI-2 ECC decoder (SM_HW_VIN_ECC)

The SoC software shall enable the MIPI CSI-2 ECC decoder by setting the MIPI_ECC_ENABLE bit in the VIN configuration registers during system initialization.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on or reset the system - Allow boot sequence and system initialization to complete - Read the VIN configuration register containing `MIPI_ECC_ENABLE` bit - Verify the bit value = 1
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111535 - [TechSaReqs] SafetyGoal 1 Requirements

MVC-99788 - Enabling the MIPI CSI-2 CRC (SM_HW_VIN_CRC)

The SoC software shall enable the MIPI CSI-2 CRC checker by setting the appropriate CRC_ERR_THRESHOLD in the VIN configuration registers during system initialization.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on or reset the system - Allow boot sequence and system initialization to complete - Read the VIN configuration register containing `CRC_ERR_THRESHOLD` bit - Verify the bit value = 1
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111535 - [TechSaReqs] SafetyGoal 1 Requirements

MVC-99789 - Read the VIN status register to check for MIPI CRC error flags

The SoC software shall periodically read the VIN status register to check for MIPI CRC error flags and MIPI ECC error flags indicating corrupted data payload and packet headers.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on or reset the system- Inject a single frame with corrupted payload CRC. - Verify that SW detects the CRC error flag. - Inject a single frame with corrupted packet header (ECC error). - Verify that SW detects the ECC error flag.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111535 - [TechSaReqs] SafetyGoal 1 Requirements

MVC-99790 - Reaction on MIPI ECC or CRC error

If the SoC software detects a MIPI ECC or CRC error in the VIN status register, and this fault is confirmed after a duration of 100ms (FDTI), the system shall complete the transition to the safe state within 200ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Inject continuous MIPI ECC or CRC errors for 150ms - Start timer at first error detection - Monitor software fault confirmation flag - Verify flag is set at 100ms after first error detection - Measure time from fault confirmation to PHY inactive - Verify this time ≤200ms
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99999 - SG_3 TSRs is related to : MVC-111535 - [TechSaReqs] SafetyGoal 1 Requirements

1.2.4 SG_4 TSRs

MVC-100092 - Check the 'VideoStreamConfiguration' configuration received from vehicle

The SoC software shall check that the 'VideoStreamConfiguration' was correctly received from vehicle at each change of data (at run-time) by checking the checksum.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Send a valid configuration message with correct checksum. - Verify software accepts the configuration and does not report any checksum error - Send the configuration message but with an invalid checksum - Verify software detects checksum mismatch and rejects the configuration
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-99800 - Reaction and safe state

If the data integrity check of 'VideoStreamConfiguration' fails and the fault is confirmed after 200ms (FDTI), the system shall complete the transition to the safe state within 100ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Start sending invalid configuration messages continuously. - Start timer at first invalid message reception. - Monitor software fault confirmation flag for 'VideoStreamConfiguration'. - Verify flag is set at 200ms - Measure time from fault confirmation to PHY inactive. - Verify this time <=100ms.

Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-100096 - The parameters stored in DRAM shall be subject to CRC-protection/control.

The parameters stored in DRAM shall be subject to CRC-protection/control.

Note: For details refer to DRAM section of this document.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	Review and inspect that CRC protection is implemented.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-100091 - Check the 'VideoStreamConfiguration' configuration written in SOW1

The SoC software shall check that the 'VideoStreamConfiguration' was correctly written at SOW1 at start-up and at each change of the data by reading back the relevant configuration registers and comparing against expected values.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	Power on system and allow initialization to complete. verify that software compares readback value with intended value. Trigger a configuration change during runtime. Confirm software accepts change only if readback matches.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-113675 - Check the establishment of the control channel and control picture

The SoC software shall check that two dedicated 'Control Channels' with their respective 'Control Pictures' (one for computer vision and one for human vision) of requested sizes are established for each active video channel at start-up and maintained at run-time.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	-Review architectural design to confirm dedicated control channels and pictures for computer vision and human vision. -Inspect source code to verify start-up initialization and run-time maintenance.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99792 - SG_4 TSRs

MVC-115165 - 'Control Picture' adjustment

When SG4 relevant parameter has changed, the SoC SW shall set up the 'Control Picture' accordingly.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	review relevant source code part. Note: the 'Control Picture' adjustmen can be tested on the system level.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99792 - SG_4 TSRs

MVC-113674 - Injection of the control picture at ISP entrance

The SoC software shall inject the respective 'Control Pictures' into the RAW Data path at the entrance of the ISP during run-time on separate, non-overlapping frames periodically every 3rd frame for computer vision and every 3rd frame for human vision.

Status	Predefined

Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	-Execute the test environment with active video streams for both vision types. -Verify that control pictures are injected on non-overlapping frames (every 3rd frame for computer vision and every 3rd frame for human vision).
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99792 - SG_4 TSRs

MVC-115144 - Validation of control picture attributes at ISP output

The SoC software shall check that the attributes (e.g., size and dimensions) of the injected 'Control Pictures' are correctly maintained and checked at the end of the ISP pipeline for both computer vision and human vision streams.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	-Stream active video data through the system pipelines. -Verify that the system successfully checks the attributes (size and dimensions) of the injected control pictures at the end of the ISP pipeline for both streams.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99792 - SG_4 TSRs

MVC-113679 - Control picture check failure.

If the attributes check of the injected 'Control Pictures' at the end of the ISP pipeline fails for either stream and the fault is confirmed after 300ms (FDTI), the system shall complete the transition to the safe state within 100ms (FRTI).

Note:

the time of checking every 3rd frame is already cosidered in the FDTI

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	-Corrupt the attributes of the control picture at the ISP output to induce a check failure. -Verify that the system detects the fault within 300ms (FDTI) and completes the transition to the safe state within 100ms (FRTI).
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99792 - SG_4 TSRs

MVC-99795 - Check configuration of Deserlializer

At system start-up and at each change of configuration data, the SoC software shall verify the configuration of deserializer by reading back the relevant configuration registers and comparing against expected values.

Status	Rejected
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on system and allow initialization to complete. - Verify that software compares readback value with intended value. - Trigger a deserializer configuration change during runtime. - Verify that software accepts change only if readback matches.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-99797 - Check configuration of SoC

At system start-up and at each change of configuration data, the SoC software software shall verify the configuration of SoC by reading back the relevant configuration registers and comparing against expected values.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on system and allow initialization to complete. - Verify that software compares readback value with intended value. - Trigger a SoC configuration change during runtime. - Verify that software accepts change only if readback matches.

Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-99799 - Check configuration of PHY

At system start-up and at each change of configuration data, the SoC software shall verify the configuration of PHY by reading back the relevant configuration registers and comparing against expected values.

Status	Rejected
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on system and allow initialization to complete. - Verify that software compares readback value with intended value. - Trigger a PHY configuration change during runtime. - Verify that software accepts change only if readback matches.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements is duplicated by : MVC-113514 - Read back of configuration write

MVC-100107 - Check configuration of power supply subsystem

At system start-up and at each change of configuration data, the SoC software shall verify the configuration of power supply subsystem by reading back the relevant configuration registers and comparing against expected values.

Status	Rejected
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on system and allow initialization to complete. - Verify that software compares readback value with intended value. - Trigger a Power supply configuration change during runtime. - Verify that software accepts change only if readback matches.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. has parent : MVC-99792 - SG_4 TSRs is duplicated by : MVC-99249 - After each write command to PMIC, the SoC SW shall read back the written value f... is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-100098 - Flash CRC verification at start-up.

At system start-up the SoC software shall verify the configuration data located in FLASH by reading the relevant configuration parameters and checking the associated CRC.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power on system and allow initialization to complete. - Verify that software compares readback value with intended value. - Trigger a Flash configuration change during runtime. - Verify that software accepts change only if readback matches.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-100097 - Flash CRC calculation and storage at shut-down.

At system shut-down the SoC software shall calculate CRC associated with configuration data and write it (together with the relevant configuration data) to the proper location at the FLASH.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI

Verification Criteria	- After system powers down, read the designated flash location where configuration data and CRC should be stored. - Calculate CRC from the stored data. - Verify recalculated CRC matches stored CRC.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95955 - The system shall ensure 'fault free' reception of 'VideoStreamConfiguration' fro... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-100099 - Flash readback verification after storage.

After having stored updated configuration data (together with the associated CRC) in the FLASH, the SoC shall verify the configuration data located in FLASH by reading back the relevant configuration parameters (together with the associated CRC) and comparing it with intended values stored in the DRAM.

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review and inspect that readback configuration verification is performed with intended values stored in the DRAM.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... has parent : MVC-99792 - SG_4 TSRs is related to : MVC-111538 - [TechSaReqs] SafetyGoal 4 Requirements

MVC-113678 - Reaction to configuration readback failure

If the readback verification of SoC configuration registers or data fails and the fault is confirmed after 200ms (FDTI), the system shall complete the transition to the safe state within 100ms (FRTI).

Status	Predefined
Assignee(s)	Ebaid, Abdelaziz
ASIL	ASIL B
Allocation	SW
Verification Criteria	-Inject a fault in the SoC configuration registers or data to provoke a readback verification failure. -Measure the time elapsed to verify that the system confirms the fault after the specified 200ms duration (FDTI). -Verify that the system successfully completes the transition to the defined safe state within 100ms (FRTI) following fault confirmation.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99792 - SG_4 TSRs

1.3 DRAM Safety Requirements

MVC-99141 - Store the WDT Configurations (SM_HW_DRAM_WDT)

The configurations (WATCH DOG_TIMER_0_CONFIG and WATCHDOG_TIMER_1_CONFIG) to enable hardware watchdog timers function shall be stored in the flash.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review the configuration map to confirm that watchdog configuration data is included in the flash at the correct locations.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99142 - Configurations of DRAM WDT (SM_HW_DRAM_WDT)

The SoC SW shall program WATCHDOG_TIMER_0_CONFIG and WATCHDOG_TIMER_1_CONFIG registers with number of cycles to expire/16.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Review that the software correctly calculates and programs the watchdog timer configuration registers with the appropriate timeout value.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111258 - Configurations of DRAM WDT (SM_HW_DRAM_WDT) is verified by : MVC-115255 - FUSA-DRAM-SM_HW_DRAM_WDT

MVC-99256 - Hardware watchdog detect a hang condition (SM_HW_DRAM_WDT)

The hardware watchdog timers shall detect a hang condition in the functional blocks of
DRAM within a maximum of 3ms from the onset of the hang.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Force a hang in the first monitored functional block - Start timer at moment hang is initiated. - Measure time until watchdog error signal asserts. - Verify the measured detection time is <=3ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is satisfied by : MVC-101052 - Hardware watchdog detect a hang condition (SM_HW_DRAM_WDT) is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99143 - Safe state transition upon DRAM WDT timeout (SM_HW_DRAM_WDT)

If the SM_HW_DRAM_WDT indicates an error, the SoC shall transition into Safe State within 200ms.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Start normal operation of system - Force a hang in the first monitored functional block - Start the timer at the moment the watchdog error signal is asserted - Stop the timer when Ethernet transmission is stopped - Verify the time constraint
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99144 - Executing SM_HW_LBIST during start-up (SM_HW_LBIST)

In order to prevent SM_HW_DRAM_WDT related latent faults, the Ambarella-Chip SW shall make sure that SM_HW_LBIST is executed during start-up.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Power cycle the system. - Verify that LBIST completed successfully - Verify that the LBIST test patterns include the logic blocks associated with SM_HW_DRAM_WDT.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99145 - Store the hardware SECDED ECC in the flash (SM_HW_HECC)

The SoC SW shall store the configuration REG_ECCH_MODE to enable hardware SECDED ECC function in the flash.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review configuration map to confirm that REG_ECCH_MODE is included in the flash at the correct location.
Verification Method	Not testable, checked by Review

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99146 - configurations of the the hardware ECC (SM_HW_HECC)

The SoC SW shall set the HW ECC engine enable bit REG_ECCH_MODE to enable the hardware SECDED ECC function for the external DRAM interface during the system startup.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review startup code to confirm that REG_ECCH_MODE is written with the correct value during DRAM controller initialization sequence.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111257 - Configurations of the the hardware ECC (SM_HW_HECC) is verified by : MVC-115256 - FUSA-DRAM-SM_HW_HECC_001

MVC-99147 - Types of DRAM (SM_HW_HECC)

The system shall use 2 LPDDR4 memories: LPDDR4 8Gbit (16x512) (ECC-Ram) and LPDDR4 16Gbit (32x512) (Data-Ram)

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Review BOM to confirm that the specified LPDDR4 devices are listed: 1) 8Gbit (16x512) device (ECC-Ram) with correct part number 2) 16Gbit (32x512) device (Data-Ram) with correct part number
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is satisfied by : MVC-101054 - Types of DRAM (SM_HW_HECC) is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99257 - Detection and correction any single-bit error (SM_HW_HECC)

The HW ECC engine shall automatically detect and correct any single-bit error within a
protected 32-byte codeword read from the external Data-RAM.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Write a known 32-byte pattern to a protected Data-RAM location. - Read back and verify data integrity. - Inject the stored data by flipping exactly one bit in the 32-byte codeword. - Perform a read of the same Data-RAM location. - Verify that the data returned matches the original pattern and confirming automatic correction.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is satisfied by : MVC-101053 - Detection and correction any single-bit error (SM_HW_HECC) is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is verified by : MVC-115258 - FUSA-DRAM-SM_HW_HECC_002

MVC-99260 - Reaction on single-bit error (SM_HW_HECC)

If single-bit error is detected and corrected, the HW shall increment
MAIN_CORRECTABLE_COUNTER OR ECC_CORRECTABLE_COUNTER on a single
bit error correction.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Write a known 32-byte pattern to a protected DRAM location. - Read back and verify data integrity. - Read ECC error counters - Inject the stored data by flipping exactly one bit in the 32-byte codeword. - Perform a read of the same DRAM location. - Verify that the data returned matches the original pattern and confirming automatic correction. - Read ECC error counters again and verify that the correctable error counter has incremented by exactly 1.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99258 - Detection double-bit error (SM_HW_HECC)

The HW ECC engine shall detect any double-bit error within a protected 32-byte
codeword read from the external Data-RAM

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Inject the stored data by flipping exactly two bits within the 32-byte codeword. - Perform a read of the same DRAM location. - Start timer at the moment the read is done - Stop timer when CEHU safety error signal asserts.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is satisfied by : MVC-101056 - Detection double-bit error (SM_HW_HECC) is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99259 - Reaction on double-bit error (SM_HW_HECC)

If an uncorrectable error (double-bit or multi-bit error) is detected, the HW ECC engine
shall assert a safety error signal to the Central Error Handling Unit (CEHU)

Status	Rejected
Assignee(s)	ElBeih, Mohamed

ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Inject an error in Data-RAM stored data by flipping exactly two bits within the 32-byte codeword. - Perform a read of the same Data-RAM location. - Start timer at the moment the read is done - Stop timer when CEHU safety error signal asserts.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is satisfied by : MVC-101055 - Reaction on double-bit error (SM_HW_HECC) is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-111784 - If an uncorrectable error (double-bit or multi-bit error) is fault is reported t...

If an uncorrectable error (double-bit or multi-bit error) is fault is reported to CEHU, the
SoC SW shall go to safe state within 200ms.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Use debugger to simulate an double-bit error reported to CEHU. - Verify the safe state is reached within the time constraint.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-115027 - If an uncorrectable error (double-bit or multi-bit error) is fault is reported t...

MVC-99148 - Store the configurations of DRAM segmentation in the flash (SM_HW_DRAM_SEGMENTATION)

The SoC SW shall store the configurations to enable the DRAM segmentation hardware function in the flash.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review configuration map to confirm that DRAM segmentation configuration data is included and is in the correct locations.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99149 - The configurations of Data RAM segmentation (SM_HW_DRAM_SEGMENTATION)

The SoC SW shall configure the Data RAM segmentation hardware registers of the SoC during the startup phase

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Check by reading back the Data RAM segmentation register configuration
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-115026 - The configurations of Data RAM segmentation (SM_HW_DRAM_SEGMENTATION) is verified by : MVC-115257 - FUSA-DRAM-SM_HW_DRAM_SEGMENTATION_001

MVC-99150 - Allocate non-overlapped memory region for DRAM segmentation (SM_HW_DRAM_SEGMENTATION)

The SoC SW shall implement DRAM segmentation configuration by allocating a minimum of one segment (one segment is 256 KB) non-overlapped memory region for each DRAM client.

Note: The System Integrator shall define and document the complete DRAM memory map for the target system, explicitly identifying all memory regions allocated to safety-critical (ASIL) and non-safety (QM) software components. This analysis shall serve as the input for the segmentation configuration.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Review source code to confirm that the allocation meets the requirements.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111259 - Allocate non-overlapped memory region for DRAM segmentation(SM_HW_DRAM_SEGMENTATION) is satisfied by : MVC-111261 - Allocate non-overlapped memory region for DRAM segmentation(SM_HW_DRAM_SEGMENTATION) is satisfied by : MVC-111262 - Allocate non-overlapped memory region for DRAM segmentation(SM_HW_DRAM_SEGMENTATION) is verified by : MVC-115260 - FUSA-DRAM-SM_HW_DRAM_SEGMENTATION_002

MVC-99151 - Ensure that each DRAM client is only permitted to access its assigned memory region(s) (SM_HW_DRAM_SEGMENTATION)

The SoC SW shall be configured to ensure that each DRAM client can only access its permitted memory region(s).

Note: The System Integrator shall configure segmentation rules to explicitly prevent all QM DRAM clients from accessing safety-critical memory regions.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- After system initialization, for each DRAM client determine an address just outside its assigned memory region. - Execute a read or write instruction to that address. - Read REG_DRAM_SEG_FAULT and verify that the corresponding fault bit is set.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111255 - Ensure that each DRAM client is only permitted to access its assigned memory region(s) (SM_HW_DRAM_SEGMENTATION)

MVC-99152 - Identify the faulty client in DRAM (SM_HW_DRAM_SEGMENTATION)

The SoC SW shall read the REG_DRAM_SEG_FAULT register to identify the faulty client and the type of violation.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW

Verification Criteria	- After system initialization, determine an address just outside its assigned region. - Execute a read or write instruction to that address. - Read REG_DRAM_SEG_FAULT and verify that the corresponding fault bit is set. - Read REG_DRAM_SEG_FAULT and identify the faulty client and the type of violation.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99153 - Safe state transition due to the faulty client (SM_HW_DRAM_SEGMENTATION)

The SoC SW shall read the REG_DRAM_SEG_FAULT register upon receiving of CEHU interrupt to check if there is segmentation access violation.
If REG_DRAM_SEG_FAULT is set, the SoC shall trigger safe state within 200ms.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- After system initialization, select an address outside its assigned region. - Execute a read or write instruction to that address. - Read REG_DRAM_SEG_FAULT and verify that the corresponding fault bit is set. - Start timer once corresponding fault bit is set. - Stop the timer when Ethernet transmission is stopped. - Verify the measured time is within limit.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111254 - Safe state transition due to the faulty client (SM_HW_DRAM_SEGMENTATION) is satisfied by : MVC-111256 - Safe state transition due to the faulty client (SM_HW_DRAM_SEGMENTATION)

MVC-99154 - Store the configurations of DRAM statistic in the flash (SM_HW_DRAM_BW_MON)

The SoC SW shall store the configuration STAT_CONTROL to enable the readout of the DRAM statistic function in the flash.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review EEPROM map to confirm that STAT_CONTROL is included in the flash and is in the correct location.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99155 - The configurations of DRAM statistic (SM_HW_DRAM_BW_MON)

The SoC SW shall enable the DRAM client traffic statistic counters by setting the STAT_CONTROL register bit during the system startup phase.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW

Verification Criteria	- Review startup code to confirm that STAT_CONTROL register is written with the correct value during system initialization sequence.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99157 - Check the DRAM statistic counter (SM_HW_DRAM_BW_MON)

The SOC SW shall disable **STAT_CONTROL** bit, check the statistic counters, and then re-enable **STAT_CONTROL** bit every 99 ms.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	Check using code inspection, that STAT_CONTROL bit is disabled before checking the statistic counters.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99158 - Evaluation of the DRAM statistic counter (SM_HW_DRAM_BW_MON)

The SoC SW shall compare and evaluate the values against the predefined application thresholds

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- During a check cycle, modify a counter to a value below its threshold. - Verify software passes the check (no fault). - Modify a counter to a value exceeding its threshold. - Verify software detects violation.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99159 - Reaction if evaluation identifies an anomaly of the DRAM statistic counter (SM_HW_DRAM_BW_MON)

If the SoC SW evaluation of the DRAM statistic counters identifies an anomaly that violates the predefined thresholds, the software shall assert a safety error signal to the Central Error Handling Unit (CEHU).

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- During a check cycle, modify a counter to a value below its threshold. - Verify software passes the check (no fault). - Modify a counter to a value exceeding its threshold. - Verify software detects violation and asserts a safety error signal to the Central Error Handling Unit
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99197 - Store the configurations of DRAM Address Stuck Test in the flash (SM_HW_ASTUCK_TEST)

The SoC SW shall store the configuration to enable the hardware DRAM Address Stuck Test function in the flash.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review configuration map to confirm that the Address Stuck Test enable configuration is included in the flash and is in the correct location.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99203 - Execution of the hardware Address Stuck Test every 100ms. (SM_HW_ASTUCK_TEST)

The SoC SW shall enable and execute the hardware Address Stuck Test (DTTE_ASTUCK_TEST_EN) every 100ms

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Read registers DTTE_ASTUCK_PERIOD, DTTE_ASTUCK_PERIOD_MSB to confirm that DTTE_ASTUCK_TEST_EN is set and that test completion is checked within time contraint.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111264 - Execution of the hardware Address Stuck Test at every startup. (SM_HW_ASTUCK_TEST) is verified by : MVC-115259 - FUSA-DRAM-SM_HW_ASTUCK_TEST_001

MVC-99204 - Configuration of the DRAM controller (SM_HW_ASTUCK_TEST)

The SoC SW shall configure the DRAM type (DRAM_TYPE register) to indicate the use of LPDDR4 memory

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Initialize the system - Read the Address Stuck Test Engine registers - Verify the DRAM_TYPE value is correct according and indicates LPDDR4 device type
Verification Method	Not testable, checked by Review

Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99205 - Configuration of the Address Stuck Test Engine (SM_HW_ASTUCK_TEST)

The SoC SW shall configure the Address Stuck Test Engine registers (DRAM_TYPE, DTTE_DUAL_DIE, DTTE_DUAL_CHANNEL, DTTE_MR15, DTTE_MR20, DTTE_EXPECTED_PATTERN, DTTE_ASTUCK_PERIOD, DTTE_ASTUCK_PERIOD_MSB) according to the specific LPDDR4 device used in the system.

Note: Refer to DDRC/DTTE Programming Reference Manual

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Initialize the system - Read the Address Stuck Test Engine registers - Verify the values are correct according to LPDDR4 device parameters.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111263 - Configuration of the Address Stuck Test Engine (SM_HW_ASTUCK_TEST) is verified by : MVC-115262 - FUSA-DRAM-SM_HW_ASTUCK_TEST_002

MVC-99206 - Monitor the status registers of the Address Stuck Test Engine (SM_HW_ASTUCK_TEST)

The SoC shall monitor the DTTE_ERR_VLD and DTTE_ASTUCK_ERR_VLD status registers as well as CEHU safety error after each execution of the Address Stuck Test and goes to safe state within 100ms if a safety error is indicated.

For more details, refer to Ambarella Safety Manual section "5.1.27 Address Stuck Test"

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Complete one Address Stuck Test execution completes with no faults. - Check the values in DTTE_ERR_VLD and DTTE_ASTUCK_ERR_VLD indicate no error. - Inject DTTE-to-CEHU error using SAFETY_ERROR_INJECTION and SAFET_MODE registers - Check the values in DTTE_ERR_VLD and DTTE_ASTUCK_ERR_VLD indicate an error. - Verify that the safety error signal is asserted and the SoC goes to safe state. - Verify time constraint.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99207 - Store the configurations of Initialization function of the DRAM BIST and Scrubbing Engine in the flash

The SoC SW shall store the configuration to enable the Initialization function of the DRAM BIST and Scrubbing Engine in the flash.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B

Allocation	HSI
Verification Criteria	- Review configuration map to confirm that the enable configuration is included in the flash and is in the correct location.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements

MVC-99208 - Configuration of the Initialization function of the DRAM BIST and Scrubbing Engine.

The boot software shall configure the DRAM BIST and Scrubbing Engine (DBSE) by setting DBSE_MODE to 0 and DBSE_BIST_PATTERN to 0x1 to initialize all addressable memory in the external DDR device before any client performs a read operation from DRAM.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Review boot code to confirm that the DBSE configuration routine is called before any code that accesses DRAM - Verify that the registers are written with the correct values (DBSE_MODE=0, DBSE_BIST_PATTERN=0x1).
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99075 - DRAM Safety Requirements is related to : MVC-105436 - [TechSaReqs] DRAM Safety Requirements is satisfied by : MVC-111253 - Configuration of the Initialization function of the DRAM BIST and Scrubbing Engine. is verified by : MVC-115261 - FUSA-DRAM-

1.4 Flash Safety Requirements

MVC-99210 - Generate CRC-32 checksums during build time (Build use case).

The build system shall generate CRC-32 for code and data sections during build time.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Build and flash the SW - Allow the flash tool to complete its verification (readback + CRC-32 compare). - Verify that the tool reports PASS and no errors. - Modify one byte in the flash file without updating the CRC-32 - Try to flash the SW again - Verify that the tool detects the mismatch and reports a failure.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements

MVC-99211 - Allocate generated checksums in write-protected region during build time (Build use case)

The build system shall allocate generated checksums in a flash region that can be write-protected.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW

Verification Criteria	- Review the linker script to identify where checksum sections are placed.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements

MVC-99212 - Store code, data, and their checksums in in write-protected region during flashing (Flashing use case)

During flashing, the flash tool shall store code, data, and their checksums in the respective flash sections and set write-protection where required.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	
Verification Method	
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements

MVC-99213 - Recalculate checksums, and compare them with stored checksums After flashing (Flashing use case)

After flashing, the flash tool shall read back the image, recalculate CRC-32, and compare them with stored CRC-32.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Build and flash the SW - Allow the flash tool to complete its verification (readback + CRC-32 compare). - Verify that the tool reports PASS and no errors.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements

MVC-99214 - Checksums error detection after flashing (Flashing use case)

If there is CRC-32 mismatch after flashing, the flash tool shall stop the flashing process and indicate a CRC-32 error.

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Modify one byte in the flash file without updating the CRC-32 - Try to flash the SW again - Verify that the tool detects the mismatch and reports a failure.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements

MVC-99215 - calculate and compare checksums with stored checksums at each startup (Startup use case)

The SoC shall load sections into RAM, calculate checksums, and compare them with stored checksums at each startup, before using safety-relevant code or data from flash.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed

ASIL	ASIL B
Allocation	SW
Verification Criteria	Verify that after startup, the checksum of safety relevant data loaded from flash is calculated and compared with stored checksum before the data is used.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements is satisfied by : MVC-111292 - Calculate and compare checksums with stored checksums at each startup (Startup use case) is satisfied by : MVC-111293 - Calculate and compare checksums with stored checksums at each startup (Startup use case) is satisfied by : MVC-111295 - Calculate and compare checksums with stored checksums at each startup (Startup use case)

MVC-99216 - Checksums error detection at startup (Startup use case)

If there is CRC-32 mismatch at startup, the SoC shall indicate a CRC-32 error and shall not start application SW.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	Change one byte in the flash area of application SW without updating the CRC-32. In startup log, check that a CRC-32 error is indicated and that application SW does not start.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements is satisfied by : MVC-111294 - Checksums error detection at startup (Startup use case) is satisfied by : MVC-111297 - Checksums error detection at startup (Startup use case)

MVC-99217 - Calculate CRC-32 checksums before storing safety-relevant data to flash after startup (Storing use case after startup)

The SoC shall calculate a CRC-32 checksum in RAM with proper integrity before storing safety-relevant data to flash

Status	Rejected
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	Verify that CRC-32 calculation is done in a SW component which is memory protected.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements

MVC-99218 - Store the data and checksum in flash (Storing use case after startup)

The SoC shall store the calculated checksum together with the actual data in flash.

Status	Approved with assumption
Assignee(s)	ElBeih, Mohamed
ASIL	ASIL B
Allocation	SW
Verification Criteria	Update the data in flash and check that update of the checksum field is carried out as well. Repeat test for all safety related sections in flash.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99074 - Flash Safety Requirements is related to : MVC-105437 - [TechSaReqs] Flash Safety Requirements is verified by : MVC-110425 - FUSA-FLASH-Checksum Storage In Flash is satisfied by : MVC-111296 - Store the data and checksum in flash (Storing use case after startup)

1.5 Deserializer Safety Requirements

MVC-100103 - At system start-up the SoC software shall configure the serializer in the tunelling mode.

At system start-up the SoC software shall configure the serializer in the tunnelling mode.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - check the suitable register (EXT11(0x383)) of the serializer if the Tun_Mode bit is set to 1m (i.e.: EXT11(0x383)::Tun_Mode == 1)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111299 - At system start-up the SoC software shall configure the serializer in the tunelling mode.

MVC-100119 - The Deserializer shall be configured in tunnelling mode.

The Deserializer shall be configured in tunnelling mode.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Inspect and confirm CFG1 is hardwired to 0V and deserializer operates in tunnel mode.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. has parent : MVC-99077 - Deserializer Safety Requirements is satisfied by : MVC-92666 - CFG1 - Input Map is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100122 - The connection between Deserializer and SoC via ERRB signal shall be established.

The connection between Deserializer and SoC via ERRB signal shall be established.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- check the connection between deserializer and SoC on the schematics
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99077 - Deserializer Safety Requirements is satisfied by : MVC-101087 - The SoC shall be able to react on ERRB signal being asserted. is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is verified by : MVC-112432 - FUSA-DESR-ERRB Connection Deserializer-SoC established

MVC-105967 - When SoC SW receives an ERRB-signal the SoC SW shall check...

When SoC SW receives an ERRB-signal the SoC SW shall check the meaning of the signal (i.e.: origin thereof) by querying the relevant registers in the deserializer via I2C bus within 50ms.

Reception of the signal shall not take longer than 50ms. When the assertion is detected, the system shall enter SafeState within 100ms.

Note: one of the reasons (besides the error in deserializer) for asserting the ERRB pin could be the SOW1 – SafeState (see: ECU_RS_372 of Functional Safety Requirement for HOST SYSTEM, Date: 2025-10-22 Revision: 17.2). This is the reason for 200ms FHTI budget in total.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- consult the relevant part of code
Verification Method	Not testable, checked by Review

Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111300 - When SoC SW receives an ERRB-signal the SoC SW shall check... is satisfied by : MVC-111302 - When SoC SW receives an ERRB-signal the SoC SW shall check... is satisfied by : MVC-111305 - When SoC SW receives an ERRB-signal the SoC SW shall check...

MVC-100123 - The connection between Deserializer and SoC via LOCK signal shall be established.

The connection between Deserializer and SoC via LOCK signal shall be established.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- check the connection between deserializer and SoC on the schematics
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is verified by : MVC-112433 - FUSA-DESR-Lock Connection Deserializer-SoC established

MVC-105968 - If SoC SW receives an LOCK-signal (assertion) the SoC SW shall...

If SoC SW receives an LOCK-signal (deassertion) the SoC SW shall check the cause of the deassertion (i.e.: origin thereof) by querying the relevant registers in the deserializer via I2C bus within 50ms. Reception of the signal shall not take longer than 50ms. When the deassertion is detected, the system shall enter SafeState within 100ms.
Note: one of the reasons (besides the error in deserializer) for asserting the ERRB pin could be the SOW1 – SafeState (see: ECU_RS_372 of Functional Safety Requirement for HOST SYSTEM, Date: 2025-10-22 Revision: 17.2). This is the reason for 200ms FHTI budget in total.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start SOW2 + SOW1 systems - interrupt GMSL connection - verify that SOW2 have reached the SafeState within 200ms (for event: "interrupt GMSL connection")
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111304 - If SoC SW receives an LOCK-signal (assertion) the SoC SW shall... is satisfied by : MVC-111306 - If SoC SW receives an LOCK-signal (assertion) the SoC SW shall... is satisfied by : MVC-111307 - If SoC SW receives an LOCK-signal (assertion) the SoC SW shall...

MVC-105969 - If the Deserializer detects FuSa problem...

If the Deserializer detects FuSa problem, it shall inform the SoC about it by asserting ERRB pin (and/or deasserting LOCK pin) within 10ms.

Note: for details refer to SF2, SF3, SF4 relevant tables found in: §8 in “Safety Manual for MAX96714/MAX96714F (GM16)” (DOCUMENT I.D. 32-100741, REVISION D)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start SOW2 system - cause error - very that the relevant pins ERRB (and/or LOCK) changed their states within 10ms after the event: error
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-105970 - The SoC shall be able to communicate with the Deserializer by means of primary- I2C bus.

The SoC shall be able to communicate with the Deserializer by means of primary- I2C bus.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- consult the schematics if connection exists
Verification Method	Not testable, checked by Review

Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100121 - Pass through I2C Channel

The SoC shall be able to communicate with the SOW1 components by means of 'primary-I2C' bus.

Note (BOSCH): "To arrive at the peripherials of the SOW1 (e.g.: imager) the main/primary (MFP9, MFP10) channel should be used from the Deserializer"

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- start the SOW2 + SOW1 system - try to connect with all components that can be reached from SoC via primary-I2C bus - verfiy that the connection was established Note: for component list and their I2C addresses refer to §3 of "Camera Interface Specification", Version 1.0, Document Number: 0220DN00JX
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111357 - Pass through I2C Channel

MVC-100223 - The SoC SW shall check REG1(0x1):: DIS_LOCAL_CC

During start-up the SoC SW shall check if REG1 (0x1):: DIS_LOCAL_CC==0b0. If the check fails the system shall log a trouble code.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - pause the boot-process - set REG1(0x1):: DIS_LOCAL_CC to 1 - resume the boot-process - observe the System logging a trouble code
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111298 - The SoC SW shall check REG1(0x1):: DIS_LOCAL_CC is satisfied by : MVC-111334 - The SoC SW shall check REG1(0x1):: DIS_LOCAL_CC

MVC-100222 - The SoC SW shall check REG1(0x1):: DIS_REM_CC

During start-up the SoC SW shall check if REG1 (0x1):: DIS_REM_CC==0b0. If the check fails the system shall log a trouble code.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- start the system - pause the boot-process - set REG1(0x1):: DIS_REM_CC to 1 - resume the boot-process - observe the System logging a trouble code
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111332 - The SoC SW shall check REG1(0x1):: DIS_REM_CC is satisfied by : MVC-111337 - The SoC SW shall check REG1(0x1):: DIS_REM_CC

MVC-106886 - The SoC SW shall check once per driver cycle if REG1(0x1):: IIC_1_EN ==0b1. If the check fails the system shall log a trouble code.

During start-up the SoC SW shall check if REG1 (0x1):: IIC_1_EN ==0b1. If the check fails the system shall log a trouble code.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI

Verification Criteria	- start the system - pause the boot-process - set REG1(0x1):: IIC_1_EN to 0 - resume the boot-process - observe the System logging a trouble code
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : ~~MVC-111326~~ - The SoC SW shall check once per driver cycle if REG1(0x1):: IIC_1_EN ==0b1. If the check fails the system shall log a trouble code. is satisfied by : ~~MVC-111336~~ - The SoC SW shall check once per driver cycle if REG1(0x1):: IIC_1_EN ==0b1. If the check fails the system shall log a trouble code.

MVC-106887 - The SoC SW shall check once per driver cycle if REG1(0x1):: IIC_2_EN ==0b1. If the check fails the system shall log a trouble code.

During start-up the SoC SW shall check if REG1 (0x1):: IIC_2_EN ==0b1. If the check fails the system shall log a trouble code.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- start the system - pause the boot-process - set REG1(0x1):: IIC_2_EN to 0 - resume the boot-process - observe the System logging a trouble code
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : ~~MVC-111324~~ - The SoC SW shall check once per driver cycle if REG1(0x1):: IIC_2_EN ==0b1. If the check fails the system shall log a trouble code. is satisfied by : ~~MVC-111330~~ - The SoC SW shall check once per driver cycle if REG1(0x1):: IIC_2_EN ==0b1. If the check fails the system shall log a trouble code.

MVC-100113 - At the system startup the SoC SW shall check if the ERRB pin in the deserializer is enabled. If not, the SoC SW shall enable ERRB pin by setting REG5 (0x5)::ERRB_EN := 0b1

At the system startup the SoC SW shall check if the ERRB pin in the deserializer is enabled. If not, the SoC SW shall enable ERRB pin by setting REG5 (0x5)::ERRB_EN := 0b1

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - pause the system before the check is performed - set REG5 (0x5)::ERRB_EN to 0b0 - resume the system - observe the REG5 (0x5)::ERRB_EN has been set to 0b1
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111328 - At the system startup the SoC SW shall check if the ERRB pin in the deserializer is enabled. If not, the SoC SW shall enable ERRB pin by setting REG5 (0x5)::ERRB_EN := 0b1

MVC-100112 - At the system startup the SoC SW shall check if the LOCK pin in the deserializer is enabled. If not, the SoC SW shall enable LOCK pin by setting REG5 (0x5)::LOCK_EN := 0b1

At the system startup the SoC SW shall check if the LOCK pin in the deserializer is enabled. If not, the SoC SW shall enable LOCK pin by setting REG5 (0x5)::LOCK_EN := 0b1

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - pause the system before the check is performed - set REG5 (0x5)::LOCK_EN to 0b0 - resume the system - observe the REG5 (0x5)::LOCK_EN has been set to 0b1
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111339 - At the system startup the SoC SW shall check if the LOCK pin in the deserializer is enabled. If not, the SoC SW shall enable LOCK pin by setting REG5 (0x5)::LOCK_EN := 0b1

MVC-100115 - When ERRB becomes disabled, the SoCSW shall...

When ERRB becomes disabled, the SoC SW shall transition the system into the SafeState within 100ms. Detection time of the ERRB-disabled-event shall be less than 100ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - disable ERRB - observe that system transitioned into safeState within the time constraints.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111321 - When ERRB becomes disabled, the SoCSW shall... is satisfied by : MVC-111338 - When ERRB becomes disabled, the SoCSW shall...

MVC-100114 - When LOCK becomes disabled, the SoC SW shall...

When LOCK becomes disabled, the SoC SW shall transition the system into the SafeState within 100ms. Detection time of the LOCK-disabled-event shall be less than 100ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - disable LOCK - observe that system transitioned into safeState within the time constraints.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111312 - When LOCK becomes disabled, the SoC SW shall... is satisfied by : MVC-111314 - When LOCK becomes disabled, the SoC SW shall...

MVC-100120 - The SoC SW shall detect the ERRB assertion from the deserializer within 100ms.

The SoC SW shall detect the ERRB assertion from the deserializer within 100ms. When the assertion is detected, the system shall enter SafeState within 100ms.

Note: one of the reasons (besides the error in deserializer) for asserting the ERRB pin could be the SOW1 – SafeState (see: ECU_RS_372 of Functional Safety Requirement for HOST SYSTEM, Date: 2025-10-22 Revision: 17.2). This is the reason for 200ms FHTI budget in total.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run system normally - Force ERRB low and maintain it for >= 100ms - Measure the time from the falling edge of ERRB to the detection. - Verify this time is <=100ms - Measure the time from the detection indicator to the moment PHY becomes inactive. - Verify this time is <=100ms
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100220 - LOCK deassertion

The SoC SW shall detect the LOCK deassertion from the deserializer within 100ms. When the deassertion is detected, the system shall enter SafeState within 100ms.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run system normally - Force LOCK pin low and maintain it for 150ms - Measure the time from the falling edge of LOCK to the detection. - Verify this time is <=100ms - Measure the time from the detection indicator to the moment PHY becomes inactive. - Verify this time is <=100ms
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100117 - The system shall ensure proper VDDIO voltage. If VDDIO within prescribed range cannot be guaranteed, the system shall enter the SafeState.

The system shall ensure proper VDDIO voltage. If VDDIO within prescribed range cannot be guaranteed, the system shall enter the SafeState.

Note: According Table 1. Recommended Operating Conditions of the Document: "MAX96714/MAX96714F/MAX96714R Single GMSL2/GMSL1 to CSI-2 Deserializer" the VDDIO shoud be in the range: 1.7V - 3.6V

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Run system normally at nominal VDDIO. - Verify no voltage faults reported and Safe State not entered. - Program power supply to ramp VDDIO down to below UV threshold - Verify under voltage faults reported and Safe State entered. - Program power supply to ramp VDDIO up above OV threshold - Verify over voltage faults reported and Safe State entered.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99077 - Deserializer Safety Requirements is satisfied by : MVC-101090 - The system shall ensure proper VDDIO voltage. If VDDIO within prescribed range cannot be guaranteed, the system shall enter the SafeState. is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100116 - The system shall ensure proper VTERM. If VTERM within prescribed range cannot be guaranteed, the system shall enter the SafeState.

The system shall ensure proper VTERM. If VTERM within prescribed range cannot be guaranteed, the system shall enter the SafeState.

Note: According Table 1. Recommended Operating Conditions of the Document: "MAX96714/MAX96714F/MAX96714R Single GMSL2/GMSL1 to CSI-2 Deserializer" the VTERM shoud be in the range: 1.14V (min) - 1.2V (typical) - 1.26V (max)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Run system normally at nominal VTERM. - Verify no voltage faults reported and Safe State not entered. - Program power supply to ramp VTERM down to below UV threshold - Verify under voltage faults reported and Safe State entered. - Program power supply to ramp VTERM up above OV threshold - Verify over voltage faults reported and Safe State entered.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99077 - Deserializer Safety Requirements is satisfied by : MVC-101089 - The system shall ensure proper VTERM. If VTERM within prescribed range cannot be guaranteed, the system shall enter the SafeState. is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100118 - The system shall check if junction temperature is less or equal to -40°C and less or equal to 125°C. If it is not the case the system shall enter the SafeState.

The system shall check if junction temperature is less or equal to -40°C and less or equal to 125°C. If it is not the case the system shall enter the SafeState within 300ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Run system normally for several monitoring cycles with temperature within range. - Verify no temperature faults are reported and Safe State is not entered. - Force the temperature reading to -50°C - Verify that the system enters safe state and PHY becomes inactive - verify that the time costraint fulfillment - Run system normally for several monitoring cycles with temperature within range. - Force the temperature reading to +130°C - Verify that the system enters safe state and PHY becomes inactive - verify that the time costraint fulfillment
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97949 - The system shall detect contrast deterioration occurred between its input and ou... satisfies : MVC-97950 - The system shall detect brightness deterioration that occurred between its input... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100125 - The primary-I2C bus shall use CRC scheme for the communication.

The primary-I2C bus shall use CRC scheme for the communication.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- consult the relevant part of the source code.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111317 - The primary-I2C bus shall use CRC scheme for the communication.

MVC-100124 - The SoC SW shall use Message Counter for primary-I2C.

The SoC SW shall use Message Counter for primary-I2C.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- consult the relevant part of the source code.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111316 - The SoC SW shall use Message Counter for primary-I2C.

MVC-100126 - During startup the SoC SW shall test if CRC scheme for primary- I2C bus is working.

During startup the SoC SW shall enable SM21 in the deserializer.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	SW
Verification Criteria	- start the system - at SoC SW prepare message with WRONG CRC attached - observe NACK response being sent from Deserializer - observe ERRB being asserted - Start the system - prepare an answer (with wrong CRC attached) from Deserializer and inject it into SoC - observe SoC SW detects the error
Verification Method	Testable, checked by Test

Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111309 - During startup the SoC SW shall test if CRC scheme for primary- I2C bus is working.

MVC-100127 - SoC SW shall check if tunneling mode is still set once per driver cycle.

At the end of the startup phase the SoC SW shall check if tunneling mode is still set. If SoC SW detects that deserializer and serializer is not in the tunneling mode the system shall log this issue and reset the system.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - pause the system before check begins - Inject and write to register 0x474 and clear bit 0 to 0 - resume the system - observe the condition being detected and log being generated - observe system being reset
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111308 - SoC SW shall check if tunneling mode is still set once per driver cycle. is satisfied by : MVC-111311 - SoC SW shall check if tunneling mode is still set once per driver cycle.

MVC-100128 - GMSL2 and I2C selection shall remain as set during startup.

When REG6::GMSL2_A becomes 0b0 or REG6::I2CSEL becomes 0b0 (i.e. configuration changes). the SoC SW shall transition the system into the SafeState within 100ms. Detection time of the change-event shall be less than 100ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- start the system - set REG6::GMSL2_A := 0b0 - observe SoC SW detecting the change within 100ms - observe SoC SW transitioning the system into SafeState within time constraints - start the system - set REG6::I2CSEL := 0b0 - observe SoC SW detecting the change within 100ms - observe SoC SW transitioning the system into SafeState within time constraints
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111303 - GMSL2 and I2C selection shall remain as set during startup. is satisfied by : MVC-111310 - GMSL2 and I2C selection shall remain as set during startup.

MVC-100214 - Coax cable select

When CTRL1::CXTP_A becomes 0b0 (i.e. configuration changes) the SoC SW shall transition the system into the SafeState. Detection time of the change-event shall be less than 100ms. The transition (taking place after detection phase) shall happen within 100ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- start the system - set CTRL1::CXTP_A := 0b0 - observe SoC SW detecting the change within 100ms - observe SoC SW transitioning the system into SafeState within time constraints
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111301 - Coax cable select is satisfied by : MVC-111319 - Coax cable select

MVC-100213 - Changes to GMSL link

When any changes to the GMSL link were performed, the SoC SW shall perfom a link reset to reinitialize the link (toggle RESET_LINK HIGH and then LOW or write 1 to RESET_ONESHOT bit of the CTRL0 (0x10) register).

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - introduce changes to GMSL link: disable I2C pass-through Channel 2 by setting REG1::IIC_2_EN := 0 - watch RESET_LINK HIGH being toggled or CTRL0::RESET_ONESHOT being set - the I2C pass-through channel becomes disabled
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111335 - Changes to GMSL link

MVC-100216 - I2C communication

The SoC SW shall enable CRC packet and message counter to monitor both write and read transactions (see: I2C for CRC + Message Counter) by doing the following:

a. Set bit CC_CRC_EN to one, of the register I2C_UART_CRC7(0x300F)
b. Set bit CC_CRC_MSGCNTR_OVR to one, of the register I2C_UART_CRC7(0x300F)
c. Set bit CC_MSGCNTR_EN to one, of the register I2C_UART_CRC7(0x300F)
d. Set bit CC_CRC_MSGCNTR_OVR to one, of the register I2C_UART_CRC7(0x300F)

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- Review source code to confirm that the I²C CRC initialization routine writes the correct value to I2C_UART_CRC7 during startup.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100215 - Communication with the serializer and imager

The SoC SW shall configure the serializer and imager via i2c connection of the deserializer

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review source code to confirm that the I²C configuration routines for both serializer and imager are called during startup
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-100218 - Read back data from deserializer

After having set a parameter in the deserializer, the SoC SW shall read back the value of the parameter and compare it with the intended (original 'set'-value). If the comparison fails, the system shall transition to the SafeState within 200ms. The transition phase follows the detection phase. The starting time for measurement of detection phase shall conicide with issuing the write command. The detection time shall be less than 100ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI

Verification Criteria	- startup the SOW2 system - set parameter in the deserializer - change the value of the pramater in the deserializer before the read attempt occurs - observe the system entering the SafeSatate within 300ms (detection + transition)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111327 - Read back data from deserializer is satisfied by : MVC-111333 - Read back data from deserializer

MVC-100219 - Read back data from SOW1-component

After having set a parameter in the SOW1-component, the SoC SW shall read back the value of the parameter and compare it with the intended (original 'set'-value). If the comparison fails, the system shall transition to the SafeState within 200ms. The transition phase follows the detection phase. The starting time for measurement of detection phase shall conicide with issuing the write command. The detection time shall be less than 100ms.

Note1: for component list and their I2C addresses refer to §3 of "Camera Interface Specification", Version 1.0, Document Number: 0220DN00JX
Note2: ATTENTION! the channels for write and read can be different depending on the parameter and its "host" (e.g.: parameter == 'exposure time', host == imager)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- startup the SOW2 system - set parameter in the SOW1-component - change the value of the paramater in the SOW1-component before the read attempt occurs - observe the system entering the SafeState within 300ms (detection + transition)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111325 - Read back data from serializer is satisfied by : MVC-111331 - Read back data from serializer

MVC-100221 - Read back data from imager

After having set a parameter in the imager, the SoC SW shall read back the value of the parameter and compare it with the intended (original 'set'-value). If the comparison fails, the system shall enter the SafeState within 200ms (detection time shall be less than 100ms).

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review source code to confirm that readback verification is implemented after every safety critical write to the imager, and that a mismatch leads to enter safe state
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : ~~MVC-111320~~ - Read back data from imager is satisfied by : ~~MVC-111329~~ - Read back data from imager

MVC-106124 - If the read command (issued by SoC SW via I2C bus) of FuSa related parametes...

If the read command (issued by SoC SW via I2C bus) of FuSa related parametes does not yield result within 50ms, the SW SoC shall transition the system into SafeState within 200ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - disturb I2C-communication - issue command to serializer (e.g. read temperature) - wait for timeout - check time costraint <=50ms - observe transition into SafeState within time constraints
Verification Method	Testable, checked by Test

Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111323 - If the read command (issued by SoC SW via I2C bus) of FuSa related parametes...

MVC-106125 - When, while performing read action via I2C bus, the SoC SW detects CRC error signaled by Deserializer, the SoC SW shall repeat the read action within 50ms

When, while performing read action via I2C bus, the SoC SW detects CRC error signaled by Deserializer, the SoC SW shall repeat the read action within 50ms.

Note (BOSCH): "To arrive at the peripherials of the SOW1 (e.g.: imager) the main/primary (MFP9, MFP10) channel should be used from the Deserializer" (ref. to MVC-100121)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HSI
Verification Criteria	- start the system - issue read command from SoC SW - inject answer with CRC fault - observe a repeated raed command being sent within time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111322 - When, while performing read action via I2C bus, the SoC SW detects CRC error signaled by Deserializer, the SoC SW shall repeat the read action within 50ms

MVC-106126 - if the re-read action via I2C bus also results in CRC error, the SoC shall transition into SafeState within 200ms (detection + reaction)

If the re-read action via I2C bus (from deserializer) also results in CRC error, the SoC shall transition into SafeState within 200ms (detection + reaction)

Note: this is a continuation of MVC-106125 (alternative path - in case of failure)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - cause two consecutive CRC-fails in the context of read request - observe that the system transitions into SafeState within time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-111315 - if the re-read action via I2C bus also results in CRC error, the SoC shall transition into SafeState within 200ms (detection + reaction)

MVC-106127 - The SoC SW shall read SOW1-parameter via 'pass-throug I2C' two times successively within 50ms.

The SoC SW shall read SOW1-parameter via 'pass-throug I2C' two times successively within 50ms.

Note:
I2C for CRC + Message Counter
Note: This section does not apply to the MAX96714R variant.
The MAX96714 provides two additional functional safety features by adding an optional CRC packet to either side
of the link and an optional message counter to monitor both write and read transactions. **Both features are available only to the deserializer's primary I2C control channel for register read/write transactions**. They are
disabled by default and must be enabled at the register level. (source: application Notes: "MAX96714/MAX96714F/MAX96714R, User Guide GMSL SerDes Applications Team Rev 0 05/2023")

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - send read command from imager - observe the command being repeated within time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-106128 - The SoC SW shall compare the results of each read-action...

The SoC SW shall compare the results of each read-action. If the comparison (for FuSa parameters) fails, the System shall transition into the SafeState within 200ms (comparison/detection + transition)

Status	Rejected
Assignee(s)	Rekas, Miroslaw

ASIL	ASIL B
Allocation	SW
Verification Criteria	- start system - perform full length (i.e. two phase) read given parameter from the imager - during re-read inject wrong value - observe that the system transitions into SafeState within time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-106129 - At startup the Deserializer shall run...

At startup the Deserializer shall run Logic BIST test, Memory BIST test and at run BIST for Serial Link test. If at least one of the tests fails, the System shall transition into SafeState within 350ms (test execution + error signaling + error detection + reaction)

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- consult the source code and make sure that conditions for triggering the tests are fulfilled.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : ~~MVC-111313~~ - At startup the Deserializer shall run...

MVC-113552 - At startup run Logic BIST test

At startup the Deserializer shall run Logic BIST test.

Note: it is a part of POST (Power-On Self-Test) and it is triggered automatically. (see: p. 49 of document: “MAX96714/MAX96714F/MAX96714R Single GMSL2/GMSL1 to CSI-2 Deserializer”, *19-101134; Rev 2; 6/23)*

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	check the capability of the HW element in the documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-113553 - Check after Logic BIST test

After Logic BIST test has been performed, the SoC SW shall check the REG_POST0 (0x3020) for the test results. If results indicate failure, the system shall transition into SafeState within 350ms (detection + reaction time).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - wait for the system to boot - observe the SoC SW checks the bits: REG_POST0[POST_RUN_LBIST] == 1 and REG_POST0[POST_LBIST_PASSED] == 1 and REG_POST0[POST_DONE] == 1 - system does NOT transition into SafeState - start the system - halt the system - manipulate the SoC SW variable responsible for REG_POST0[POST_RUN_LBIST] and REG_POST0[POST_LBIST_PASSED] and REG_POST0[POST_DONE] to indicate an error - observe System transitioning to SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-114524 - Check after Logic BIST test

MVC-113555 - At startup run Memory BIST test

At startup the Deserializer shall run Memory BIST test.

Note: it is a part of POST (Power-On Self-Test) and it is triggered automatically. . (see: p. 49 of document: “MAX96714/MAX96714F/MAX96714R Single GMSL2/GMSL1 to CSI-2 Deserializer”, 19-101134; Rev 2; 6/23)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	check the capability of the HW element in the documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements

MVC-113556 - Check after Memory BIST test

After Memory BIST test has been performed, the SoC SW shall check the REG_POST0 (0x3020) for the test results. If results indicate failure, the system shall transition into SafeState within 350ms (detection + reaction time).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - wait for the system to boot - observe the SoC SW checks the bits: REG_POST0[POST_RUN_MBIST] == 1 and REG_POST0[POST_MBIST_PASSED] == 1 and REG_POST0[POST_DONE] == 1 - system does NOT transition into SafeState - start the system - halt the system - manipulate the SoC SW variable responsible for REG_POST0[POST_RUN_MBIST] and REG_POST0[POST_MBIST_PASSED] and REG_POST0[POST_DONE] to indicate an error - observe System transitioning to SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : MVC-114523 - Check after Memory BIST test

MVC-106130 - During startup the SoC SW shall enable reporting of line-fault interrupt.

During startup the SoC SW shall enable reporting of line-fault interrupt.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - disturb the connection between SOW1 and SOW2 - observe ERRB signal being asserted - observe that system transitions into SafeState
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99077 - Deserializer Safety Requirements is related to : MVC-105438 - [TechSaReqs] Deserializer Safety Requirements is satisfied by : ~~MVC-111318~~ - During startup the SoC SW shall enable reporting of line-fault interrupt.

1.6 Ethernet PHY Safety Requirements

Revision before Rework: /wiki/40_System_Allocation_Requirements/Technical_Safety_Concept?sidebar=approvals&revision=687679

1.6.1 Constraints (operating conditions)

MVC-113529 - When System HW detects that the PHY temperature exceeds limits, the System HW shall transition into the SafeState within 350ms (realization: power down)

When System HW detects that the PHY temperature exceeds limits, the System HW shall transition into the SafeState within 350ms (realization: power down)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW

Verification Criteria	- switch off the system - bring the PHY above the upper temperature limit - switch on the system - observe the system transitions into SafeState or remains into SafeState - verify time constraint was satisfied - switch off the system - bring the PHY below the lower temperature limit - switch on the system - observe the system transitions into SafeState or remains into SafeState - verify time constraint was satisfied - start the system - wait for the system to boot - bring the PHY above the upper temperature limit - observe system transition into SafeState - verify time constraint was satisfied - start the system normally (nominal temperature) - wait for the system to boot - bring the PHY below the lower temperature limit - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-108304 - Constraints (operating conditions) is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113530 - Operating conditions consistent with the PHY documentation

The SoC shall use the PHY under operating conditions consistent with the documentation (such as the data sheet) and the PHY mission profile.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	review the documentation of PHY and SOW2
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-108304 - Constraints (operating conditions) is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

1.6.2 Configuration

MVC-108301 - Set config enable.

Before starting configuration of PHY, the SoC SW shall set the CONFIG_ENABLE bit to 1 (to unlock for reconfiguration).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Set CONFIG_ENABLE register - Attempt to change a configuration register - Read back the register value and check it has been changed.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99583 - After configuration is complete, the SoC SW shall reset the CONFIG_ENABLE bit to...

After configuration is complete, the SoC SW shall reset the CONFIG_ENABLE bit to protect the configuration from unintended modifications during runtime.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Reset CONFIG_ENABLE register - Attempt to change a configuration register - Read back the register value and check it is not changed.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements is satisfied by : MVC-111290 - After the safety-critical configuration is complete, the software driver shall r...

MVC-99627 - The SoC shall use the interrupt INT_N to react to the failures in...

The SoC software shall use the interrupt INT_N to react to the failures in Ethernet PHY.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	check the HW design.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111789 - No JTAG enabled

The PHY shall NOT provide JTAG interface

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	check/review HW documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111790 - The PHY shall provide RGMII-ID (RXC) mode of operation/interface.

The PHY shall provide RGMII-ID (RXC) mode of operation/interface.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	QM
Allocation	HW
Verification Criteria	check/review HW documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111791 - The PHY shall provide interrupt interface to the SoC. The SoC shall be able to r...

The PHY shall provide interrupt interface to the SoC. The SoC shall be able to receive interrupts from the PHY via this interface.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	check/review HW documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111792 - The PHY shall provide reset interface to the SoC. The SoC shall be able to reset...

The PHY shall provide reset interface to the SoC. The SoC shall be able to reset the PHY via this interface.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B

Allocation	HW
Verification Criteria	check/review HW documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111793 - The PHY shall provide MDIO interface for configuration and status read purposes....

The PHY shall provide MDIO interface for configuration and status read purposes. The SoC shall be able to configure the PHY via this interface.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	check/review HW documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

6. ~~The PHY shall go through a complete power up/power down cycle (A full power up/power down cycle including VDDA_AO, asserting RST_N or wake-up from DEVICE_DEEP_SLEEP) within the latent fault tolerant time interval (L-FTTI), which is assumed to be 24 hrs. -> during the startup phase the SoC shall reset the PHY -> this is requirement on VOLVO from SOW2~~

MVC-111794 - After PHY startup, the SoC shall activate interrupt signaling within PHY by conf...

After PHY startup, the SoC shall activate interrupt signaling within PHY by configuring (set active) relevant IRQ sources at PHY by doing:

a. Set PORT_IRQ_ENABLES[PORT1_IRQ] :=1
    i. Set PORT_FUNC_IRQ_ENABLES[SERDES_MII_IRQ] :=1
        1. Set PORT_LEVEL_IRQ_ENABLES[EPHY_CLK_WARN] :=1
        2. Set PORT_LEVEL_IRQ_ENABLES[EPHY_LF] :=1
        3. Set PORT_LEVEL_IRQ_ENABLES[TXC_CLK_WARN] :=1
        4. Set PORT_LEVEL_IRQ_ENABLES[RXC_CLK_WARN] :=1
    ii. Set PORT_FUNC_IRQ_ENABLES[INFRA_IRQ] :=1
        1. Set INFRA_IRQ_ENABLES[FIFO_ERROR]:=1
        2. Set INFRA_IRQ_ENABLES[RGMII_ID_LOCK_ERROR]:=1
    iii. Set PORT_FUNC_IRQ_ENABLES[EPHY_IRQ] :=1
        1. Set EPHY_CAT_IRQ_ENABLES[EPHY_CONFIG_REGXS_IRQ]:=1
            a. Set PHY_CONFIG_REGXS_IRQ_ENABLES[RECONFIG_DENIED] :=1
            b. Set PHY_CONFIG_REGXS_IRQ_ENABLES[INIT_ERROR] :=1
            c. Set PHY_CONFIG_REGXS_IRQ_ENABLES[SMI_ERROR] :=1
        2. Set EPHY_CAT_IRQ_ENABLES[EPHY_FUNCTIONAL_IRQ]:=1
            a. Set EPHY_FUNCTIONAL_IRQ_ENABLE[LINK_AVAILABLE_EVENT] := 1
b. Set PORT_IRQ_ENABLES[GLOBAL_IRQ] :=1
    i. GLOBAL_CAT_IRQ_ENABLE[CONFIG_REGXS_IRQ]:=1
        1. Set CONFIG_REGXS_IRQ_ENABLE[SMI_ERROR]:= 1
    ii. GLOBAL_CAT_IRQ_ENABLE[OP_CONDITIONS_IRQ]:=1
        1. Set OP_CONDITIONS_IRQ_ENABLE[TEMP_WARN]:=1
        2. Set OP_CONDITIONS_IRQ_ENABLE[OSC_WARN]:=1
        3. Set OP_CONDITIONS_IRQ_ENABLE[VDDA_TRX_SUPPLY_WARN]:=1
        4. Set OP_CONDITIONS_IRQ_ENABLE[VREGA_SUPPLY_WARN]:=1
        5. Set OP_CONDITIONS_IRQ_ENABLE[VREGD_SUPPLY_WARN]:=1
        6. Set OP_CONDITIONS_IRQ_ENABLE[VDDIO_SUPPLY_WARN]:=1
        7. Set OP_CONDITIONS_IRQ_ENABLE[CORE_SUPPLY_WARN]:=1
        8. Set OP_CONDITIONS_IRQ_ENABLE[AO_SYSTEM_SUPPLY_WARN]:=1

Note: The PHY shall report potential safety hazard by means of interrupt signal to the SoC.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- configure the relevant registers - read back if the configuration was performed correctly
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-108303 - Configuration is related to : ~~MVC-99626~~ - If a clock failure is reported as indicated by REFCLK_WARN field in ALWAYS_ACC... is related to : ~~MVC-99637~~ - The SoC shall use the interrupt INT_N to react to the failures in... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

~~8. When the PHY reports the detection of a (potential) safety hazard (by means of interrupt signal to the SoC), the SoC shall detect the reported fault and transition into SafeState within 300 ms.(detection + reaction) -> obsolete via reqs. 25 - 43~~

MVC-111795 - The System HW shall connect VDDA_TRX to VREGA_OUT.

The System HW shall connect VDDA_TRX to VREGA_OUT.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- check if there is 0 Ohm connection between VREGA_OUT and VDDA_TRX
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-108303 - Configuration is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

1.6.3 Boot-Up

MVC-113532 - While in the PHY boot up phase, the SOW2 (i.e.: SoC SW) shall remain in the SafeState).

While in the PHY boot up phase, the SOW2 (i.e.: SoC SW) shall remain in the SafeState).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - observe the system remains in the SafeState as long as the PHY is in the boot up process
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111212 - Boot-Up is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99298 - The Ethernet PHY (TJA1104 or TJA1121) shall execute a Built-In Self-Test (BIST) sequence...

The Ethernet PHY (TJA1104 or TJA1121) shall execute a Built-In Self-Test (BIST) sequence during every device startup, for both device-level and port-level self-test checks.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- After power up, read the BIST status registers - Verify that the BIST has completed.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-111212 - Boot-Up is satisfied by : MVC-101091 - BIST execution is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99575 - If all BIST checks successfully completed, the Ethernet PHY shall automatically set t...

If all BIST checks successfully completed, the Ethernet PHY shall automatically set the DEVICE_STATUS[FUSA_STATUS] bit to '1' and trigger FUSA_PASS_IRQ signal.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- After power up, read the BIST status registers - Verify that the BIST has completed and FUSA_PASS_IRQ signal was asserted - Verify that the FUSA_STATUS bit is set to 1.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-111212 - Boot-Up is satisfied by : MVC-101094 - BIST successful is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99577 - If any BIST check fails, the Ethernet PHY shall report the specific failure by settin...

If any BIST check fails, the Ethernet PHY shall report the specific failure by setting the corresponding Latent Fault bit in the related mechanism status registers (xx_LATENT_FAULT)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw

ASIL	ASIL B
Allocation	HW
Verification Criteria	- After power up, read the BIST status registers - Verify that the BIST has completed and FUSA_PASS_IRQ signal was asserted - Verify that the FUSA_STATUS bit is set to 1. - configure the PHY to force a failure - Reset the PHY to trigger a new BIST run. - Immediately after reset read DEVICE_STATUS[FUSA_STATUS] - Verify that the bit remains 0 - Immediately after reset read ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] - Verify that the bit remains 0 - Immediately after reset, read the latent fault status registers. - Verify that the bit corresponding to the forced failure is set to 1.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-111212 - Boot-Up is satisfied by : MVC-101093 - BIST unsuccessful is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99576 - The SoC shall confirm the successful completion of the Ethernet PHY startup BIST by eit...

The SoC SW shall confirm the successful completion of the Ethernet PHY startup BIST by:
Waiting for and receiving the FUSA_PASS_IRQ interrupt, and subsequent checking that ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] is set to 1.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - observe signal INT_N being set to low and confirm that ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] is set to 1
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... has parent : MVC-111212 - Boot-Up is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements is satisfied by : MVC-111291 - The SoC shall confirm the successful completion of the Ethernet PHY startup BIST by eit...

MVC-99579 - If the SoC detects a BIST failure of Ethernet PHY (indicated by FUSA_STATUS remaining '0' after...

If the SoC software detects a BIST failure of Ethernet PHY (indicated by FUSA_STATUS remaining '0' after timeout or by reading latent fault bits), the SoC shall enter safe state within FHTI (300ms)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- After power up, read the BIST status registers - Verify that the BIST has completed and FUSA_PASS_IRQ signal was asserted - Verify that the FUSA_STATUS bit is set to 1. - Reset the PHY to trigger a new BIST run. - stop execution before BIST - write to the PHY's latent fault status register - Start a timer - Resume execution - Check PHY pin INT_N is asserted low - Verify the time constraints
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-111212 - Boot-Up is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113531 - 'Link established' signal an indicator for transitioning into 'normal operation' state

At the end of the boot-up phase when the SoC has received 'link established' interrupt (EPHY_FUNCTIONAL_IRQ_SOURCE[LINK_AVAILABLE_EVENT] == 1 AND PHY_STATUS[LINK_AVAILABLE] == 1), the SOW2 (i.e.: SoC SW) shall switch to 'normal operation' state.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - observe 'link established' interrupt - observe SoC SW detercting and reacting to this interrupt (e.g.: making transition to other state in the State Machine)
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111212 - Boot-Up is related to : ~~MVC-99636~~ - If a link failure is reported as indicated by LINK_AVAILABLE_EVENT in... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

1.6.4 Run-Time

MVC-111796 - The System HW shall detect when VDDA_AO is outside of the range (3.135V; 3.465V) value and will enter a safe state (by the cutting off the VDDA_AO power supply) within 50 ms.

The System HW shall detect when VDDA_AO is outside of the range (3.135V; 3.465V) value and will enter a safe state (by the cutting off the VDDA_AO power supply) within 50 ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start the system - modify the VDDA_AO above the upper limit value - observe system transition into SafeState - verify time constraint was satisfied - start the system - modify the VDDA_AO below the lower limit value - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111802 - The System HW shall detect when VDDIO is outside of the range (1.65V; 1.95V) value and will enter a safe state (by the cutting off the VDDIO power supply) within 50 ms.

The System HW shall detect when VDDIO is outside of the range (1.65V; 1.95V) value and will enter a safe state (by the cutting off the VDDIO power supply) within 50 ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start the system - modify the VDDIO above the upper limit value - observe system transition into SafeState - verify time constraint was satisfied - start the system - modify the VDDIO below the lower limit value - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111803 - When SoC receives lost-link interrupt (INT_N asserted and EPHY_FUNCTIONAL_IRQ_MS...

When SoC receives lost-link interrupt (INT_N asserted and EPHY_FUNCTIONAL_IRQ_MSTATUS [LINK_AVAILABLE_EVENT] == 1 and PHY_STATUS[LINK_AVAILABLE] == 0), the SoC sets a 100BASE-T1 link-recovery timeout period to 200ms and maintains normal operation during the 100BASE-T1 link-recovery timeout. The interrupt reception and link recovery timer setup shall take no longer than 50ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW

Verification Criteria	- start the system - disrupt the Automotive Ethernet link - observe INT_N signalled by PHY - observe SoC checks the information w.r.t. EPHY_FUNCTIONAL_IRQ_MSTATUS [LINK_AVAILABLE_EVENT] == 1 and PHY_STATUS[LINK_AVAILABLE] == 0 - observe the link-recovery timeout being set to 200ms - verify that the time constraint is met - observe normal operation (sending/receiving) for 200ms
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99636~~ - If a link failure is reported as indicated by LINK_AVAILABLE_EVENT in... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111804 - If SoC receives the link recovery interrupt (INT_N asserted AND EPHY_FUNCTIONAL_...

If SoC receives the link recovery interrupt (INT_N asserted AND EPHY_FUNCTIONAL_IRQ_MSTATUS [LINK_AVAILABLE_EVENT] == 1 AND PHY_STATUS[LINK_AVAILABLE] == 1) within the recovery time, SoC shall reset and stop the timer (timer inactive). The SoC shall continue its normal operation. The interrupt reception and link recovery timer stoppage/clearing shall take no longer than 50ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- bring system into 'lost-link interrupt condition' (see: MVC-111803) - within recovery time re-establish Automotive Ethernet link - observe INT_N signalled by PHY - observe SoC checks the information w.r.t. EPHY_FUNCTIONAL_IRQ_MSTATUS [LINK_AVAILABLE_EVENT] == 1 and PHY_STATUS[LINK_AVAILABLE] == 1 - observe SoC stops and resets the timer - verify that the time constraint is met - observe SoC normal operation
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99636~~ - If a link failure is reported as indicated by LINK_AVAILABLE_EVENT in... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111805 - If SoC does not receive the recovery interrupt within the timeout period, the So...

If SoC does not receive the recovery interrupt within the timeout period, the SoC shall transition into the SafeState. The whole process of link failure detection + reaction shall not take longer than 100ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- bring system into 'lost-link interrupt condition' (see: MVC-111803) - wait for link-recovery time to elapse - observe System transitions into SafeState - verify that the time constraint is met
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99636~~ - If a link failure is reported as indicated by LINK_AVAILABLE_EVENT in... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111806 - SoC shall provide ‘read/write-parameter’ routine/service towards Vehicle.

SoC shall provide ‘read/write-parameter’ routine/service towards Vehicle.

Note: This requirement is related to requirement imposed on vehicle w.r.t. writing and reading-back parameters

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- select a parameter (e.g. image size, flip) - send parameter to SoC - read back the parameter - verify that the read value is equal to the original (write parameter) value
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111807 - SleepMode deactivation before entering SafeState

If the SoC shall transition into SafeState, the SoC shall deactivate SleepMode of the PHY by doing the following:

a. Verify that no active transition between states takes place: if PHY_STATE[BUSY] == 1, then wait until PHY_STATE[BUSY] == 0

b. If PHY_STATE[BUSY] == 0, read PHY_STATE[BASIC_STATE] and verify if the state machine is in one of the following states: SLEEP, SLEEP_SILENT,

SEND_IDLE_OR_DATA_SLEEP_CHECK

i. If state machine is in one of the above states, then set WAKE_SLEEP_CONTROL[SLEEP_REJECT] := 1 and do the following:

1. Wait until PHY_STATE[BUSY] == 0 and subsequently
2. Verify that state machine in NOT in one of the states above

ii. Otherwise: do nothing

c. Set PHY_CONTROL[CONFIG_ENABLE]:=1 (to unlock the configuration)

d. Set WAKE_SLEEP_CONFIG[FUNCTION_SELECT]:= 0b00 (deactivate the Wake-Sleep functionality)

e. Set PHY_CONTROL[CONFIG_ENABLE]:=0 (to lock the configuration

Note: this requirement does not affect HW-realization of SafeState.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- bring PHY into 'TC10_Ready' state - send SLEEP_REQUEST to SOW2 - inject fault into the system that will result in non-HW SafeState realization (e.g. disable video stream or delay image in the system) - verify that SleepMode is deactivated - verify that system transitioned into non-HW SafeState
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111809 - SoC shall make sure that SleepMode stays deactivated throughout the duration of...

SoC shall make sure that SleepMode stays deactivated throughout the duration of the SafeState, by polling WAKE_SLEEP_CONFIG[FUNCTION_SELECT] every 300ms, If WAKE_SLEEP_CONFIG[FUNCTION_SELECT] != 0b00 the SoC shall trigger reset of the system within 100ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- provoke non-HW SafeState of SOW2 - verify that WAKE_SLEEP_CONFIG[FUNCTION_SELECT] is polled every 300ms - restart the System - provoke non-HW SafeState of SOW2 - modify SoC variable representing WAKE_SLEEP_CONFIG[FUNCTION_SELECT] to be NOT equal 0b00 - observe System to go to HW-SafeState - verify that the time constraint is met
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111810 - disabling BIST generator

If the SoC shall transition into SafeState After PHY boot up, the SoC shall disable the BIST generator by doing the following:

a. Set DEVICE_CONTROL[GLOBAL_CONFIG]:=1

b. Set PORT_FUNC_ENABLES[BIST_ENABLE] := 0

c. Set DEVICE_CONTROL[GLOBAL_CONFIG]:=0

Note: this requirement does not affect HW-realization of SafeState.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- bring PHY into 'TC10_Ready' state - send SLEEP_REQUEST to SOW2 - inject fault into the system that will result in non-HW SafeState realization (e.g. disable video stream or delay image in the system) - verify that BIST generator is deactivated - verify that system transitioned into non-HW SafeState
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-111811 - The SoC shall make sure that BIST generator stays deactivated throughout the duration of the SafeState

The SoC shall make sure that BIST generator stays deactivated throughout the duration of the SafeState, by polling PORT_FUNC_ENABLES[BIST_ENABLE] every 300ms. If PORT_FUNC_ENABLES[BIST_ENABLE] != 0 the SoC shall trigger reset of the system within 100ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw

ASIL	ASIL B
Allocation	SW
Verification Criteria	- provoke non-HW SafeState of SOW2 - verify that PORT_FUNC_ENABLES[BIST_ENABLE] is polled every 300ms - restart the System - provoke non-HW SafeState of SOW2 - modify SoC variable representing PORT_FUNC_ENABLES[BIST_ENABLE] to be NOT equal 0 - observe System to go to HW-SafeState - verify that the time constraint is met
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-112504 - FUSA_PASS_IRQ check after boot up

After PHY boot up, the SoC shall await an interrupt. When the interrupt is received the ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] and DEVICE_STATUS[FUSA_STATUS] bits shall be checked. If ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] == 1 and DEVICE_STATUS[FUSA_STATUS] == 1 then do nothing, else the SoC shall transition into SafeState.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - in the SoC SW change the variables containing values of the registers: ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] (variable1) and DEVICE_STATUS[FUSA_STATUS] (variable2) to: --- variable1 := 0, variable2 := 0, or --- variable1 := 0, variable2 := 1, or --- variable1 := 1, variable2 := 0 - observe the SoC transitioning into SafeSate
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-112505 - FUSA_PASS_IRQ check after power release

After power release (release of RST_N) the SoC shall await an interrupt. When the interrupt is received the ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] and DEVICE_STATUS[FUSA_STATUS] bits shall be checked. If ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] == 1 and DEVICE_STATUS[FUSA_STATUS] == 1 then do nothing, else the SoC shall transition into SafeState.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - in the SoC SW change the variables containing values of the registers: ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] (variable1) and DEVICE_STATUS[FUSA_STATUS] (variable2) to: --- variable1 := 0, variable2 := 0, or --- variable1 := 0, variable2 := 1, or --- variable1 := 1, variable2 := 0 - observe the SoC transitioning into SafeSate Note: basically the same test as in MVC-112504
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-112506 - FUSA_PASS_IRQ check after DEVICE_DEEP_SLEEP state

After wake-up from DEVICE_DEEP_SLEEP state, the SoC shall await an interrupt. When the interrupt is received the ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] and DEVICE_STATUS[FUSA_STATUS] bits shall be checked. If ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] == 1 and DEVICE_STATUS[FUSA_STATUS] == 1 then do nothing, else the SoC shall transition into SafeState.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - in the SoC SW change the variables containing values of the registers: ALWAYS_ACCESSIBLE[FUSA_PASS_IRQ] (variable1) and DEVICE_STATUS[FUSA_STATUS] (variable2) to: --- variable1 := 0, variable2 := 0, or --- variable1 := 0, variable2 := 1, or --- variable1 := 1, variable2 := 0 - observe the SoC transitioning into SafeSate Note: basically the same test as in MVC-112504
Verification Method	Testable, checked by Test

Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-112546 - XYZ_IRQ_ENABLE check

After PHY was configured, the SoC shall read relevant IRQ-bits of the relevant 'XYZ_IRQ_ENABLE' registers (the ones that has been set active) and compare them with the target values in the SoC. If any discrepancy is detected, the SoC shall transition the system into SafeState within 300ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - observe the SoC SW preforms the check of the relevant XYZ_IRQ_ENABLE bits - start the system - before SoC SW performs the check, change values of the reference variables (the once that mirror the current/valid/expected value of the XYZ_IRQ_ENABLE register's bits) - observe System going into SafeState - verify time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-112547 - No active interrupt present

If the SoC detects interrupt, the SoC shall search for the source of the interrupt in relevant bits of the relevant <<XYZ>>_IRQ_MSTATUS registers and ALWAYS_ACCESSIBLE register. If no active interrupt source was found, the system shall enter SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - bring INT_N signal to low - observe SoC SW initiates screening of relevant bits in the <<XYZ>>_IRQ_MSTATUS and ALWAYS_ACCESSIBLE registers - observe System going into SafeState - verify time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-112548 - clear the source of the interrupt

When SoC detects an interrupt, the SoC shall clear the source of the interrupt by doing the following:

a. Write 0b1 to bit in question (source of the interrupt) of the relevant *_IRQ_SOURCES or ALWAYS_ACCESSIBLE register

b. After the bit was cleared, the SoC SW shall check if the clearing process was successful by reading back the bit value (expected value 0b0, 'write one to clear' bit)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - provoke RECONFIG_DENIED interrupt by: --- reading PHY_PARAMETERS[LINK_AVAILABLE_HYST_TIME] --- writing the read value to the PHY_PARAMETERS[LINK_AVAILABLE_HYST_TIME] - observe INT_N going to low - observe SoC SW performs search for the source of the interrupt - observe SoC SW performs clearing (by writing 0b1) of the EPHY_CONFIG_REGXS_IRQ_SOURCES[RECONFIG_DENIED] bit - observe SoC SW performs read back of the EPHY_CONFIG_REGXS_IRQ_SOURCES[RECONFIG_DENIED] - verify that EPHY_CONFIG_REGXS_IRQ_SOURCES[RECONFIG_DENIED] contains 0b0 value
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113093 - If the SoC detects EPHY_CLK_WARN, the system shall transition into SafeState within 340ms (detection + reaction).

If the SoC SW detects EPHY_CLK_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW

Verification Criteria	- start the system - wait for the system to boot - modify SoC internal SW variable responsible for EPHY_CLK_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113102 - If the SoC detects EPHY_LF, the system shall transition into SafeState within 340ms (detection + reaction).

If the SoC SW detects EPHY_LF, the SoC SW shall transition into SafeState within 340ms (detection + reaction).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for EPHY_LF - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113137 - If the SoC detects TXC_CLK_WARN, the system shall transition into SafeState within 340ms (detection + reaction).

If the SoC SW detects TXC_CLK_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for TXC_CLK_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113138 - If the SoC detects RXC_CLK_WARN, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects RXC_CLK_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for RXC_CLK_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113139 - If the SoC detects FIFO_ERROR, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects FIFO_ERROR, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for FIFO_ERROR - observe system transition into SafeState - verify time constraint was satisfied

Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113140 - If the SoC detects RGMII_ID_LOCK_ERROR, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC detects RGMII_ID_LOCK_ERROR, the system shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for RGMII_ID_LOCK_ERROR - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113141 - If the SoC detects RECONFIG_DENIED, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects RECONFIG_DENIED, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for RECONFIG_DENIED - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113482 - If the SoC detects INT_ERROR, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects INT_ERROR, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for INT_ERROR - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113485 - If the SoC detects EPHY_CONFIG_REGXS_IRQ[SMI_ERROR], the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects EPHY_CONFIG_REGXS_IRQ[SMI_ERROR], the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for EPHY_CONFIG_REGXS_IRQ[SMI_ERROR] - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

~~23.If the SoC detects LINK_AVAILABLE==0, the system shall transition into SafeState within 340ms (detection + reaction) -> covered in req. 12 and 13 (link recovery)~~

MVC-113490 - If the SoC detects CONFIG_REGXS_IRQ_SOURCES[SMI_ERROR], the system shall transition into SafeState within 340ms (detection + reaction)

24. If the SoC SW detects CONFIG_REGXS_IRQ_SOURCES[SMI_ERROR], the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for CONFIG_REGXS_IRQ_SOURCES[SMI_ERROR] - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113496 - If the SoC detects TEMP_WARN, the system shall record the temperature. If the temperature exceeds the maximum limit, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects TEMP_WARN, the SoC SW shall record the temperature. If the temperature exceeds the maximum limit, the system shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for TEMP_WARN so that the value exeeds the maximum limit of the temperature - observe the temperature has beed recorded - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113497 - If the SoC detects OSC_WARN, the system shall transition into SafeState within 340ms (detection + reaction) (realization: reset)

If the SoC SW detects OSC_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction) (realization: reset)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for OSC_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : ~~MVC-99626~~ - If a clock failure is reported as indicated by REFCLK_WARN field in ALWAYS_ACC... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113499 - If the SoC detects VDDA_TRX_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects VDDA_TRX_SUPPLY_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for VDDA_TRX_SUPPLY_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113505 - If the SoC detects VREGA_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects VREGA_SUPPLY_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for VREGA_SUPPLY_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113508 - If the SoC detects VREGD_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects VREGD_SUPPLY_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for VREGD_SUPPLY_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113509 - If the SoC detects VDDIO_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects VDDIO_SUPPLY_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for VDDIO_SUPPLY_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113510 - If the SoC detects CORE_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction) (realization: reset)

If the SoC SW detects CORE_SUPPLY_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction) (realization: reset)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for CORE_SUPPLY_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113511 - If the SoC detects AO_SYSTEM_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction)

If the SoC SW detects AO_SYSTEM_SUPPLY_WARN, the SoC SW shall transition into SafeState within 340ms (detection + reaction)

Status	Predefined

Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - modify SoC internal SW variable responsible for AO_SYSTEM_SUPPLY_WARN - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99580~~ - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113512 - The System HW shall detect when VDD_CORE is outside of the range (1.045V; 1.55V) value and will enter a safe state (by the cutting off the VDD_CORE power supply) within 50 ms

The System HW shall detect when VDD_CORE is outside of the range (1.045V; 1.55V) value and will enter a safe state (by the cutting off the VDD_CORE power supply) within 50 ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- modify the VDD_CORE above the upper limit value - observe system transition into SafeState - verify time constraint was satisfied - modify the VDD_CORE below the lower limit value - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113513 - The SoC shall use a reference clock that is independent of the PHY reference clock.

The SoC shall use a reference clock that is independent of the PHY reference clock.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	review the schematics and verify the independecy of clocks.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113514 - Read back of configuration write

The SoC SW shall read back and verify every configuration write. If the readback does not match the intended configuration, it will redo the configuration and readback. If that still fails, the SoC SW shall transition into SafeState. The whole process write-readback-write-readback + detection + transition shall take no longer than 350ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - perform configuration write (configuration variable) - observe SoC SW reads back the written configuration variable - modify SoC internal SW variable responsible for the configuration variable - observe system redo the configuration and readback - modify SoC internal SW variable responsible for the configuration variable - observe system transition into SafeState - verify time constraint was satisfied
Verification Method	Testable, checked by Test
Linked Work Items	duplicates : ~~MVC-99799~~ - Check configuration of PHY has parent : MVC-111210 - Run-Time is related to : ~~MVC-99582~~ - The SoC software shall implement a read-back verification procedure for every sa... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113515 - Read back of safety status register

The SoC SW shall read back and verify every safety status register read. If the readback does not match the initial read, it will redo the readback and take the majority vote.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system- wait for the system to boot - set INT_N to low - observe SoC SW reads the status registers - observe SoC SW reads back the status registers - modify the read back entry (SoC internal SW variable) - observe SoC SW reads back the status registers - observe system takes majority vote Note: for time constraint refer to relevant status (fault signaling) requirement
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : ~~MVC-99584~~ - If the readback does not match the intended configuration, the SoC SW shall redo... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113518 - VDD_CORE check with highest priority

After any interrupt, the SoC SW shall check the VDD_CORE UVD/OVD interrupt flag and handle it before proceeding with normal operation.

Status	Rejected
Assignee(s)	
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - disable mechanism described in MVC-113512 - provoke VDD_CORE UVD - observe the SoC SW checking VDD_CORE as first - observe system transition into SafeState - start the system - disable mechanism described in MVC-113512 - provoke VDD_CORE OVD - observe the SoC SW checking VDD_CORE as first - observe system transition into SafeState
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113521 - OSC_WARN check with highest priority

After any interrupt, the SoC SW shall check the OSC_WARN interrupt flag and handle it before proceeding with normal operation.

Note: time constraints will be checked in MVC-113497

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - provoke OSC_WARN - observe the SoC SW checking OSC_WARN as one of the first signals
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113522 - Lost-link interrupt check with highest priority

After any interrupt, the SoC SW shall check the lost-link interrupt flag(s) and handle them before proceeding with normal operation.

Note: for lost-link handling refer to: MVC-111803, MVC-111804, MVC-111805

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - provoke lost-link - observe the SoC SW checking lost-link interrupt as one of the first signals
Verification Method	Testable, checked by Test

Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-113525 - Temperature interrupt check with highest priority

After any interrupt, the SoC SW shall check the temperature flag(s) and handle them before proceeding with normal operation.

Note: time constraints will be checked in MVC-113496

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - provoke temperature warning - observe the SoC SW checking TEMP_WARN interrupt as one of the first signals
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111210 - Run-Time is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

1.6.5 Miscellaneous

MVC-113526 - The SoC shall implement timestamping on the application level for the videostream.

The SoC shall implement timestamping on the application level for the videostream.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - observe images at the output contain a timestamp
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111209 - Miscellaneous

MVC-113527 - The SoC SW (MAC transmission) shall apply FCS to transmitted Ethernet frames.

The SoC SW (MAC transmission) shall apply FCS to transmitted Ethernet frames.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - observe Ethernet frames at the output contain a FCS
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111209 - Miscellaneous

MVC-113528 - The SoC SW (receiving MAC) shall detect corrupted received frames via the FCS check and discard them.

The SoC SW (receiving MAC) shall detect corrupted received frames via the FCS check and discard them.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - send corrupted Eternet frame - observe the system detects and discards the frame
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-111209 - Miscellaneous

1.6.6 Requirements to Volvo

ACTTENTION: for PHY related requirements on Vehicle see 4 - Requirements Towards Vehicle

~~1. The vehicle shall put the SOW2 system through a complete power up/power down cycle (A full power up/power down cycle including VDDA_AO, asserting RST_N or wake-up from DEVICE_DEEP_SLEEP) within the latent fault tolerant time interval (L-FTTI), which is assumed to be 24 hrs. (what is the L-FTTI on the SOW2 level?) -> see: MVC-113533~~

~~2. Each 'set-parameter' command sent from Vehicle to SOW2, shall be followed by read-back command (issued by Vehicle) concerning the parameter relevant to the 'set-parameter' command within application specific time (so that safety goals will not be violated, refer to Vehicle Safety Concept) -> see: MVC-100095~~

~~3. After having received the parameter value, the Vehicle shall compare the received value with the original value (from 'set-parameter' command). If the values do not match, the Vehicle shall react accordingly to the Safety Concept on the vehicle level. -> see: MVC-100095~~

~~4. The Vehicle is able to handle frame loss or to implement appropriate safety measures against frame loss (for example a message counter) when required by the application. (e.g. readback, timestamping) -> MVC-113534~~

1.6.7 Communication with PHY

~~write/read back~~

1.6.8 FuSa Checks by PHY

~~The PHY shall implement undervoltage and overvoltage detection of VDD_CORE. If PHY detects VDD_CORE is out of range, PHY shall set INT_N to LOW.~~

1.6.9 System reaction on FuSa error

~~SoC reacts on INT_N.~~

~~Register auswerten, SafeState anleiten~~

1.6.9.1 Communication with PHY

~~Acknowledgement of INT_N (attention: hier is different than normal write!)~~

~~configure in~~

~~enable intrrupt in OP_CONDITIONS_IRQ_ENABLES:~~

~~1. TEMP_WARN -> SM11~~

~~2. OSC_WARN -> SM4~~

~~7. CORE_SUPPLY_WARN ->SM3~~

~~enable intrrupt in EPHY_FUNCTIONAL_IRQ_ENABLES[LINK_AVAILABLE_EVENT]~~

~~POLARITY_CONFLICT~~
~~TRAINING_FAILURE~~
~~TRANSMIT_FAILURE~~
~~WAKE_SLEEP_EVENT~~
~~LINK_AVAILABLE_EVENT~~
~~LINK_STATUS_EVENT~~

~~enable intrrupt in SGMII_IRQ_ENABLE[RX_LINK_STATUS_EVENT_ENABLE]~~

~~if int_n occurred, check the bits in the registers:~~

~~OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] == 1 && ALWAYS_ACCESSIBLE [CORE_SUPPLY_WARN] == 1 && (CORE_SUPPLY_STATUS[OV_EVENT] == 1 || CORE_SUPPLY_STATUS[UV_EVENT] == 1) -> SM3~~

~~OP_CONDITIONS_IRQ_SOURCES[OSC_WARN] == 1 && ALWAYS_ACCESSIBLE [REFCLK_WARN] == 1 && (PORT_LEVEL_IRQ_SOURCES[EPHY_CLK_WARN] == 1 || PORT_LEVEL_IRQ_SOURCES[TXC_CLK_WARN]== 1) -> SM4~~

~~EPHY_FUNCTIONAL_IRQ_SOURCES[LINK_AVAILABLE_EVENT] && PHY_STATUS[LINK_AVAILABLE] == 0 -> SM5~~

~~OP_CONDITIONS_IRQ_SOURCES[TEMP_WARN] && (TEMP_STATUS[HIGH_TEMP_EVENT] == 1 || TEMP_STATUS[HIGH_TEMP_NOW] == 1) -> SM11~~

~~SGMII_IRQ_SOURCE[RX_LINK_STATUS_EVENT] && SGMII_ADVANCED_STATUS[LINK_STATUS] == 0 -> SM_6~~

~~to acknowledge the int_n~~

~~write 1 to source bit of the interrupt of the EPHY_FUNCTIONAL_IRQ_SOURCES or OP_CONDITIONS_IRQ_SOURCES depending on the event that triggered interrupt.~~

MVC-99578 - The SoC software shall continuously monitor the ALWAYS_ACCESSIBLE register for a...

The SoC software shall continuously monitor the ALWAYS_ACCESSIBLE register for active fault indications from the Ethernet PHY.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system for monitoring cycles. - Verify that the software does not report any fault - Confirm that the ALWAYS_ACCESSIBLE register reads as 0 - Inject fault by writing to the ALWAYS_ACCESSIBLE register - Monitor the software and verify the fault detection
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99580 - Upon detection of an active fault of Ethernet PHY, the SoC shall assert and hold...

Upon detection of an active fault of Ethernet PHY, the SoC shall hold Ethernet PHY in reset (RST_N = LOW) within FHTI (300ms)

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system for monitoring cycles. - Verify that the software does not report any fault - Confirm that the ALWAYS_ACCESSIBLE register reads as 0 - Inject fault by writing to the ALWAYS_ACCESSIBLE register - Monitor the software and verify the fault detection - Start timer once detection - Monitor PHY to become inactive - Verify it is <= 300 ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... relates to : MVC-113093 - If the SoC detects EPHY_CLK_WARN, the system shall transition into SafeState within 340ms (detection + reaction). relates to : MVC-113102 - If the SoC detects EPHY_LF, the system shall transition into SafeState within 340ms (detection + reaction). relates to : MVC-113137 - If the SoC detects TXC_CLK_WARN, the system shall transition into SafeState within 340ms (detection + reaction). relates to : MVC-113138 - If the SoC detects RXC_CLK_WARN, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113139 - If the SoC detects FIFO_ERROR, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113140 - If the SoC detects RGMII_ID_LOCK_ERROR, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113141 - If the SoC detects RECONFIG_DENIED, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113482 - If the SoC detects INT_ERROR, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113485 - If the SoC detects EPHY_CONFIG_REGXS_IRQ[SMI_ERROR], the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113490 - If the SoC detects CONFIG_REGXS_IRQ_SOURCES[SMI_ERROR], the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113496 - If the SoC detects TEMP_WARN, the system shall record the temperature. If the temperature exceeds the maximum limit, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113497 - If the SoC detects OSC_WARN, the system shall transition into SafeState within 340ms (detection + reaction) (realization: reset) relates to : MVC-113499 - If the SoC detects VDDA_TRX_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113505 - If the SoC detects VREGA_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113508 - If the SoC detects VREGD_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113509 - If the SoC detects VDDIO_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction) relates to : MVC-113510 - If the SoC detects CORE_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction) (realization: reset) relates to : MVC-113511 - If the SoC detects AO_SYSTEM_SUPPLY_WARN, the system shall transition into SafeState within 340ms (detection + reaction) has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99582 - The SoC software shall implement a read-back verification procedure for every sa...

The SoC software shall implement a read-back verification procedure for every safety-critical configuration write to the Ethernet PHY.

Safety-critical configuration encompasses the following signals: CORE_SUPPLY_WARN, REFCLK_WARN, LINK_AVAILABLE, FUSA_PASS_IRQ, RX_STATUS.

Note: for details refer to other requirements in this chapter.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run the system normally - Verify that all read-backs match the written values - Inject Fault for a Specific Register - Resume execution. - Verify that the software performs the read-back and detect the mismatch.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... relates to : MVC-113514 - Read back of configuration write has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99581 - The SoC software shall compare the read-back value with the intended value.

The SoC software shall compare the read-back value with the intended value.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system normally - Verify that all read-backs match the written values - Inject Fault for a Specific Register - Resume execution. - Verify that the software performs the read-back and detect the mismatch.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99584 - If the readback does not match the intended configuration, the SoC SW shall redo...

If the readback does not match the intended configuration, the SoC SW shall redo the configuration and readback. If the readback still fails, the SoC SW shall transition into SafeState. The whole process shall not take longer than 350ms (detection + transition).

Note: no reception of readback values is cosidered as not matching intended configuration and should be taken into acount under 350ms limit.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run the system normally - Verify that all read-backs match the written values - Inject Fault for a Specific Register - Resume execution. - Verify that the software performs the read-back and detect the mismatch and corrected. - Verify and confirm the retry logic (exactly one retry) and the notification mechanism is implemented.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... relates to : MVC-113515 - Read back of safety status register has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99601 - The Ethernet PHY shall implement undervoltage and overvoltage detection of VDDA...

The Ethernet PHY shall implement undervoltage and overvoltage detection of VDDA_AO, the always-on supply of the Ethernet PHY.

Status	Predefined

Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	Review schematic to verify UV and OV are implemented in circuitry.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-95936 - The system shall ensure fault free 'fault free' reception of 'VideoStreamConfigu... satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-95955 - The system shall ensure 'fault free' reception of 'VideoStreamConfiguration' fro... has parent : MVC-108308 - Communication with PHY is satisfied by : MVC-101095 - VDDA_AO undervoltage and overvoltage monitoring is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99600 - If VDDA_AO is outside its operating range, the Ethernet PHY HW shall assert and hold...

The System electronics shall detect undervoltage and overvoltage condition of VDDA_AO. If VDDA_AO is outside its operating range (deviates by more than 10 % from the specified nominal value), the System electronics shall hold Ethernet PHY in reset (RST_N = LOW) within FHTI (10ms).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Run system with VDDA_AO at nominal voltage. - Program the power supply to drop or exceed "VDDA_AO" the voltage threshold - Start time immediately - Monitor PHY to become inactive as RST_N goes low. - Verify it is <= 10 ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-108308 - Communication with PHY is satisfied by : MVC-101097 - VDDA_AO undervoltage and overvoltage reaction is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99604 - The system shall implement undervoltage and overvoltage detection of VDDIO, t...

The System electronics shall detect undervoltage and overvoltage condition of VDDIO. If VDDIO is outside its operating range (deviates by more than 10 % from the specified nominal value), the System electronics shall hold Ethernet PHY in reset (RST_N = LOW) within FHTI (10ms).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Set VDDIO to nominal voltage - Decrease VDDIO voltage below the 10%threshold - verify that PHY is reset (RST_N = LOW) within the time constraints - Set VDDIO to nominal voltage - Increase VDDIO voltage above the 10%threshold - verify that PHY is reset (RST_N = LOW) within the time constraints
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-95936 - The system shall ensure fault free 'fault free' reception of 'VideoStreamConfigu... satisfies : MVC-95955 - The system shall ensure 'fault free' reception of 'VideoStreamConfiguration' fro... has parent : MVC-108308 - Communication with PHY is satisfied by : MVC-101100 - The system shall implement undervoltage and overvoltage detection of VDDIO, t... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99603 - If VDDIO is outside its operating range, the system shall assert and hold...

If VDDIO is outside its operating range (deviates by more than 10 % from the specified nominal value), the System shall hold Ethernet PHY in reset (RST_N = LOW) within FHTI (10ms).

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW

Verification Criteria	- Set VDDIO to nominal voltage - Verify PMIC status registers show no VDDIO faults. - Decrease VDDIO voltage below threshold - Verify PMIC status registers show UV VDDIO faults. - Once detection start time immediately - Monitor PHY to become inactive. - Verify it is <= 10 ms. - Increase VDDIO voltage above threshold - Verify PMIC status registers show OV VDDIO faults. - Once detection start time immediately - Monitor PHY to become inactive. - Verify it is <= 10 ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-108308 - Communication with PHY is satisfied by : MVC-101099 - If VDDIO is outside its operating range, the system shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99606 - The system shall implement undervoltage and overvoltage detection of VDD_CORE

The PHY shall implement undervoltage and overvoltage detection of VDD_CORE.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Set VDD_CORE to nominal voltage - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 0 - Decrease VDD_CORE voltage - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 1 - Set VDD_CORE to nominal voltage - write 1 to OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 0 - Increase VDD_CORE voltage - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 1
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-95936 - The system shall ensure fault free 'fault free' reception of 'VideoStreamConfigu... satisfies : MVC-95955 - The system shall ensure 'fault free' reception of 'VideoStreamConfiguration' fro... has parent : MVC-108308 - Communication with PHY is satisfied by : MVC-101102 - The system shall implement undervoltage and overvoltage detection of VDD_CORE is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99605 - If VDD_CORE is outside its operating range, the system shall assert and hold...

If VDD_CORE is outside its operating range, the system shall hold Ethernet PHY in reset (RST_N = LOW) within FHTI (300ms).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Set VDD_CORE to nominal voltage - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 0 - Decrease VDD_CORE voltage - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 1 - Once detection start time immediately - Monitor PHY to become inactive. - Verify it is <= 300ms. - Set VDD_CORE to nominal voltage - write 1 to OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] to clear the error - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 0 - Increase VDD_CORE voltage - Verify PHY status register OP_CONDITIONS_IRQ_SOURCES[CORE_SUPPLY_WARN] is equal 1 - Once detection start time immediately - Monitor PHY to become inactive. - Verify it is <= 300 ms.
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-108308 - Communication with PHY is satisfied by : MVC-101101 - If VDD_CORE is outside its operating range, the system shall assert and hold... is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99626 - If a clock failure is reported as indicated by REFCLK_WARN field in ALWAYS_ACC...

If a clock failure is reported as indicated by REFCLK_WARN field in ALWAYS_ACCESSIBLE register (801Fh), the SoC software shall hold Ethernet PHY in reset (RST_N = LOW) within FHTI (300ms).

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system normally - Inject the fault by writing to register 801Fh - Start the time immediately - Verify that the software detects the fault (Observe the RST_N pin) - Verify the time from injection to the falling edge of RST_N is <= 300ms
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... relates to : MVC-111794 - After PHY startup, the SoC shall activate interrupt signaling within PHY by conf... relates to : MVC-113497 - If the SoC detects OSC_WARN, the system shall transition into SafeState within 340ms (detection + reaction) (realization: reset) has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99637 - The SoC shall use the interrupt INT_N to react to the failures in...

The SoC software shall use the interrupt INT_N to react to the failures in Ethernet PHY link failures. The interrupt shall be enabled via EPHY_FUNCTIONAL_IRQ_ENABLES register (80A1h).

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Enable interrupt (EPHY_FUNCTIONAL_IRQ_ENABLES register) - Run the system normally with no faults injected. - Verify that the ISR INT_N is not executed - Inject a PHY fault - Verify that ISR INT_N is executed
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... relates to : MVC-111794 - After PHY startup, the SoC shall activate interrupt signaling within PHY by conf... has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

MVC-99636 - If a link failure is reported as indicated by LINK_AVAILABLE_EVENT in...

If a link failure is reported as indicated by LINK_AVAILABLE_EVENT in EPHY_FUNCTIONAL_IRQ_SOURCES (80A0h), the SoC software shall hold Ethernet PHY in reset (RST_N = LOW) within FHTI (300ms).

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system normally - Inject the fault by writing to register (80A0h) - Start the time immediately - Verify that the software detects the fault (Observe the RST_N pin) - Verify the time from injection to the falling edge of RST_N is <= 300ms
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... relates to : MVC-113531 - 'Link established' signal an indicator for transitioning into 'normal operation' state relates to : MVC-111803 - When SoC receives lost-link interrupt (INT_N asserted and EPHY_FUNCTIONAL_IRQ_MS... relates to : MVC-111805 - If SoC does not receive the recovery interrupt within the timeout period, the So... relates to : MVC-111804 - If SoC receives the link recovery interrupt (INT_N asserted AND EPHY_FUNCTIONAL_... has parent : MVC-108308 - Communication with PHY is related to : MVC-105439 - [TechSaReqs] Ethernet PHY Safety Requirements

1.7 PMIC Safety Requirements

MVC-99235 - The buck converter MPQ4572 shall monitor input voltage (Vin) of PMIC MPQ70165F

The buck converter MPQ4572 shall monitor input voltage (Vin) of PMIC MPQ70165F

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	Checked by review
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-98471 - Buck Converter is satisfied by : MVC-98501 - PMIC Buck Converter is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-105396 - The input voltage (Vin) of PMIC MPQ70165F shall be monitored

The input voltage (Vin) of PMIC MPQ70165F shall be monitored. The system HW shall cut Vin supply in case of Vin falling below 'UV threshold' or exceeding 'OV threshold' within a time that prevents permanent damage to the PMIC MPQ70165F and is below 350ms.

Definitions:
UV threshold == 3.0V
OV threshold == 5.3V

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Start up the system - set Vin to the nomina voltage - verfify that the system is working - set voltage < 3.0V - verify that the Vin (of PMIC MPQ70165F) drops down to 0V +/- tolerance (TDB) - check time constraints are fulfilled - Start up the system - set Vin to the nomina voltage - verfify that the system is working - set voltage > 5.3V - verify that the Vin (of PMIC MPQ70165F) drops down to 0V +/- tolerance (TDB) - check time constraints are fulfilled
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114719 - FUSA-PMIC-VINUVOV — PMIC Input Voltage (Vin) Monitoring and Cutoff Verified

MVC-99236 - The buck converter shall set PG at high (for rising Vout) if

The buck converter shall set PG at high (for rising Vout) if
Vout > UV threshold AND
Vout < OV threshold

Definitions:
UV threshold == 90% * Vnominal
OV threshold == 116% * Vnominal

Status	Rejected

Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Set Vout to 80% of nominal voltage. - Verify PG pin is low - Increase Vout from 80% of nominal voltage to above the 'UV threshold' and within (90% - 116%) * Vnominal range - Verify that PG transitions from low to high. - Increase Vout from the previous value (i.e. lying within (90% - 116%) * Vnominal) to over the 'OV threshold' - Verify that PG transitions from high to low.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. satisfies : MVC-95936 - The system shall ensure fault free 'fault free' reception of 'VideoStreamConfigu... satisfies : MVC-95955 - The system shall ensure 'fault free' reception of 'VideoStreamConfiguration' fro... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101103 - The buck converter shall set PG at high if Vout > 90% * Vnominal AND Vout < 108%... is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-99238 - The buck converter shall set PG at low if

The buck converter shall set PG at low if

Vout < 84% * Vnominal (which indicates under-voltage condition) OR

Vout > 116% * Vnominal (which indicates over-voltage condition)

Note:

Currently nominal output voltage of buck converter (Vnominal) = 5V

It should be checked if it is possible to change Vnominal to 4.7V so that buck converter can indicate OV if it rises to 5.5V (1.16% * Vnominal)

Above 5.5V, the operation of PMIC safety mechanisms cannot be guaranteed.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Set Vout to nominal voltage. - Verify PG pin is high - Decrease Vout from nominal down to below the UV threshold - Verify that PG transitions from high to low. - Increase Vout from nominal to over the OV threshold - Verify that PG transitions from high to low.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101105 - The buck converter shall set PG at low if Vout < 84% * Vnominal (which indicates... is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-105400 - If the Vin is oustside the defined range...

If the Vin is oustside the defined range the system shall bring the Buck Converter Switching Node Voltage of all used Outputs to 0V and assert (pulling low) ALERT# and PG# lines within 50ms.

Definitions:

'defined range ' == defined as voltage values between 'UV threshold' and 'OV threshold' (including the threshold values)

UV threshold == 3.0V

OV threshold == 5.3V

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW

Verification Criteria	- Set Vin to nominal voltage. - Verify that Buck Converter Switching Node Voltage of all used Outputs have their nominal voltage and ALERT# is high and PG# is high - Decrease Vin from nominal down to below the UV threshold - Verify that Buck Converter Switching Node Voltage of all used Outputs dropped to 0V and ALERT# is low and PG# is low - Verify time constraints. - Set Vin to nominal voltage. - Verify that Buck Converter Switching Node Voltage of all used Outputs have their nominal voltage and ALERT# is high and PG# is high - Increase Vin from nominal down to above the OV threshold - Verify that Buck Converter Switching Node Voltage of all used Outputs dropped to 0V and ALERT# is low and PG# is low - Verify time constraints.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114720 - FUSA-PMIC-VINFAULTREACT — PMIC Buck Switching-Node Collapse and ALERT#/PG# Assertion on Out-of-Range Vin

MVC-99240 - The SoC SW shall set the dedicated ALERT# register (ALERT_PG_FORCE (0xC0) bit 1)...

The SoC SW shall set the dedicated ALERT# register (ALERT_PG_FORCE (0xC0) bit 1) to pull it low at startup

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review the startup sequence to confirm that a write to ALERT_PG_FORCE (0xC0) with bit 1 = 1.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-115192 - Inhibit the PMIC PG# or ALERT# while performing toggling tests

While in the boot up phase the System HW shall inhibit the PMIC's PG# and ALERT# signals to propagate towards SoC's POR_L pin.

Note: this is to enable the PG# or ALERT# pin toggling tests as required by Safety Manual (see: § 5.2.1 PG# Power-Up Test, § 5.2.2 ALERT# Power-Up Test in document: MPQ70165FS Safety Manual Rev. 0.3 5/7/2024) and to avoid unwanted SoC reset during the test

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	Boot up system Verify that during toggling test the signal does not arrive at POR_L pin of the SoC
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-115193 - Activation of the forwarding of the PG# and ALERT# towards SoC's POR_L pin.

After PG#/ALERT# toggling tests have been completed, the SoC shall activate the forwarding of the PG# and ALERT# towards SoC's POR_L pin.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	• Start the system • After toggling tests have been completed, observe the activation signal is sent from SoC.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-115194 - Inhibition of the PG#/ALERT# towards SoC

System HW shall allow for inhibition of the PG#/ALERT# signals towards SoC.

Note: This solution shall be fail safe (i.e. in case of the fault of the inhibition control, the system be able to receive PG#/ALERT# signals)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	Review HW design.

Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-105401 - The system shall toggle the ALERT# at power up to ensure the ALERT# function is working properly in case of relevant failures.

The SoC SW shall toggle the ALERT# line of MPQ70165FS' (by toggling bit 1 (called ALERT#_FORCE_ASSERT) of ALERT_PG_FORCE (0xC0) register) at power up phase to ensure the ALERT# function is working properly in case of relevant failures. Time between assertion of the bit and detection of the ALERT# line change shall be less than 50ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - observe ALERT# having value high - observe ALERT#_FORCE_ASSERT bit having value low - set ALERT#_FORCE_ASSERT bit to High - observe ALERT# having value low - observer time constraints - set ALERT#_FORCE_ASSERT bit to Low - observe ALERT# having value high - observer time constraints Note: at power up phase, it will be necessary to wait until the i2c interface has been initilized
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111275 - The system shall toggle the ALERT# at power up to ensure the ALERT# function is working properly in case of relevant failures. is satisfied by : MVC-111286 - The system shall toggle the ALERT# at power up to ensure the ALERT# function is working properly in case of relevant failures. is verified by : MVC-113862 - FUSA-PMIC-ALERTFRC — ALERT# Force Function Verified at Power-Up

MVC-99239 - During startup, if the SoC SW detects that ALERT# pin is not properly pulled low...

During startup, if the SoC SW detects that ALERT# pin is not properly pulled low, the system shall remain in its safe state and no normal operation is allowed. The detection of the ALERT# signal shall take less than 50ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - force ALERT# signal to low - observe SoC detecting ALERT# signal having low value -> system remains/transition into SafeState - observe time constraints - force ALERT# signal to high - observe SoC detecting ALERT# signal having high value -> system remains in the operational mode - observe time constraints
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111274 - During startup, if the SoC SW detects that ALERT# pin is not properly pulled low… is satisfied by : MVC-111287 - During startup, if the SoC SW detects that ALERT# pin is not properly pulled low…

MVC-99242 - The SoC SW shall set the dedicated PG# register (ALERT_PG_FORCE (0xC0) bit 0) to...

The SoC SW shall set the dedicated PG# register (ALERT_PG_FORCE (0xC0) bit 0) to pull it low at startup

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Review the startup sequence to confirm that a write to ALERT_PG_FORCE (0xC0) with bit 0= 1.
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-105402 - The system shall toggle the PG# line of MPQ70165FS' ...

The SoC SW shall toggle the PG# line of MPQ70165FS' (by toggling bit 0 (called PG_FORCE_ASSERT) of ALERT_PG_FORCE (0xC0) register) at power up to ensure the PG# function is working properly in case of power good failures. Time between assertion of the bit and detection of the PG# line change shall be less than 50ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - observe PG# having value high - observe PG_FORCE_ASSERT bit having value low - setPG_FORCE_ASSERT bit to High - observe PG# having value low - observer time constraints - set PG_FORCE_ASSERT bit to Low - observe PG# having value high - observer time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-110542 - MVC-105402-TC-Verify_PG#_startup_safety_path_test using_PG_FORCE_ASSERT is satisfied by : MVC-111273 - The system shall toggle the PG# line of MPQ70165FS' ... is satisfied by : MVC-111277 - The system shall toggle the PG# line of MPQ70165FS' ... is verified by : MVC-113860 - FUSA-PMIC-PGFRC PG# Force Function Verified at Power-Up

MVC-99241 - During startup, if the SoC SW detects that PG# pin is not properly pulled low, t...

During startup, if the SoC SW detects that PG# pin is not properly pulled low, the system shall remain in its safe state and no normal operation is allowed. The detection of the PG# signal shall take less than 50ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- start the system - force PG# signal to low - observe SoC detecting PG# signal having low value -> system remains/transition into SafeState - observe time constraints - force PG# signal to high - observe SoC detecting PG# signal having low high -> system remains in the operational mode - observe time constraints
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111276 - During startup, if the SoC SW detects that PG# pin is not properly pulled low, t... is satisfied by : MVC-111280 - During startup, if the SoC SW detects that PG# pin is not properly pulled low, t...

MVC-99244 - The SoC SW shall use CRC for I2C communication (Tx and Rx) with PMIC using the f...

The SoC SW shall use CRC for I2C communication (Tx and Rx) with PMIC using the following parameters:

- Number of bits to check: 32 bits
- Polynomial: 0x83
- Polynomial: C(x) = x^8 + x^2 + x + 1

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Localize the implementation file - Verify that the CRC was implemented properly
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95936 - The system shall ensure fault free 'fault free' reception of 'VideoStreamConfigu... satisfies : MVC-95955 - The system shall ensure 'fault free' reception of 'VideoStreamConfiguration' fro... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111284 - The SoC SW shall use CRC for I2C communication (Tx and Rx) with PMIC using the f...

MVC-99243 - If the CRC check fails in PMIC, the PMIC shall:

If the CRC check fails in PMIC, the PMIC shall:

- Pull ALERT# low (configured via ALERT_MAPPING (0xC4), bit [2])
- Pull PG low (after a certain delay time, PG goes back to high, as configured via PG_MAPPING (0xC6), bit [2]; the PG delay time is set via PG_ALERT_DELAY, bits [1:0])

The detection and reaction shall take less than 1ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Perform normal I²C transactions with the PMIC - Inject CRC Error - Verify that ALERT# low - Verify that PG# goes low and stays low as programmed value PG_ALERT_DELAY - Verify time constraints
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101108 - CRC check unsuccessful is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114729 - FUSA-PMIC-CRCFAULT — PMIC Self-Detected CRC Fault → ALERT#/PG Reaction Verified

MVC-107742 - initialization of I2C communication in PMIC

During startup the SoC SW shall configure the I2C SoC to PMIC communication check with the following parameters:

- set bit no. 2 (PMBUS_CRC_ALERT#_MAPPING) of the register ALERT#_MAPPING (C4h) to zero (i.e. ALERT#_MAPPING[2] := 0b0) to enable ALERT# signaling
- set bit no. 2 (PMBUS_CRC_PG_MAPPING) of the register PG_MAPPING (C6h) to zero (i.e. PG_MAPPING[2] := 0b0) to enable PG signaling
- set bits no. 0 - 3 (PG_DELAY and ALERT#_DELAY) of the register PG_ALERT#_DELAY (C8h) to zero (PG_ALERT#_DELAY[0-4] := 0b000) to set the delay to 0
- set bit no. 0 (PEC_REQ) of the register SAFETY_CONFIG (C2h) to one (SAFETY_CONFIG[0] := 1) to enable packet error checking

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start the system - wait till the system boots up completely - verify the values of the bits/registers given above
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111279 - Initialization of I2C communication in PMIC is verified by : MVC-114975 - FUSA-PMIC-I2CINIT — SoC I²C Communication Check Initialization Verified

MVC-99237 - The SoC SW shall transition to safe state within FRTI (200ms) if it detects tha...

While in runtime the SoC SW shall transition to safe state within 200ms if it detects that PG# or ALERT# are pulled low, The 200ms is the time required for signal detection (on the SoC pin) and transition phase.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI

Verification Criteria	-Run system normally - Verify PG# and ALERT# are high. - Force the PG# pin low - Measure the time from PG falling edge to PHY inactivity. - Verify this time is <=200 ms. -Run system normally - Verify PG# and ALERT# are high. - Force the ALERT# pin low - Measure the time from ALERT# falling edge to PHY inactivity. - Verify this time is <=200 ms. -Run system normally - Verify PG# and ALERT# are high. - Force the PG# pin low - Measure the time from PG falling edge to PHY inactivity. - Verify this time is <=200 ms. - force ALERT# pin low - verify that the state of the system hasn't change -Run system normally - Verify PG# and ALERT# are high. - Force the ALERT# pin low - Measure the time from ALERT# falling edge to PHY inactivity. - Verify this time is <=200 ms. - force PG# pin low - verify that the state of the system hasn't change
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... relates to : MVC-112674 - internal CR: additional voltage supervision on PMIC outputs has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111278 - The SoC SW shall transition to safe state within FRTI (200ms) if it detects tha... is satisfied by : MVC-111281 - The SoC SW shall transition to safe state within FRTI (200ms) if it detects tha... is verified by : MVC-113863 - FUSA-PMIC-PGSAFE SoC Safe-State Transition on PG# Low During Runtime is verified by : MVC-113872 - FUSA-PMIC-ALERTSAFE SoC Safe-State Transition on ALERT# Low During Runtime

MVC-115187 - The System HW shall route PMIC's PG signal to SoC's POR_L input

The System HW shall route PMIC's PG signal to SoC's POR_L input.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	• Power up system • Wait for system to boot up • Provoke safety error by injecting errors in the I2C communication from SoC towards PMIC • Observe signal at POR_L input to go low
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-115188 - The System HW shall route PMIC's ALERT signal to SoC's POR_L input

The System HW shall route PMIC's ALERT signal to SoC's POR_L input.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	• Power up system • Wait for system to boot up. • Provoke safety error by injecting errors in the I2C communication from SoC towards PMIC • Observe signal at POR_L input to go low
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-115189 - If SoC HW receives the POR_L signal...

When SoC HW receives the POR_L signal, the SoC HW shall transition into SafeState within 100ms (detection + reaction).

Status	Predefined

Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	• Power up system • Wait for system to boot up. • Provoke safety error by injecting errors in the I2C communication from SoC towards PMIC • Observe System arriving at SafeState (no communication towards vehicle) • Verify time constraint.
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-115190 - The SoC SW shall activate continuous OTP CRC

The SoC SW shall activate continuous OTP CRC in PMIC by setting the bit MFR_OTP_MEM_CHK[MFR_OTP_CRC_CHK_EN] := 1

Note: this SM is not enabled by default (see: p. 67 of document: MPQ70165FS Rev. 1.0 11/3/2025)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	• Boot up system • Verify the MFR_OTP_MEM_CHK[MFR_OTP_CRC_CHK_EN] == 1 • Verify the MFR_OTP_MEM_CHK[MFR_OTP_MEM_CHK] == 1
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-115191 - The SoC SW shall activate continuous Register CRC Check

The SoC SW shall activate continuous Register CRC Check in PMIC by setting the bit MFR_OTP_MEM_CHK[MFR_MEM_CRC_CHK_EN] := 1

Note: this SM is not enabled by default (see: p. 67 of document: MPQ70165FS Rev. 1.0 11/3/2025)

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	• Boot up system • Verify the MFR_OTP_MEM_CHK[MFR_MEM_CRC_CHK_EN] == 1 • Verify the MFR_OTP_MEM_CHK[MFR_OTP_MEM_CHK] == 1
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements

MVC-99250 - If the CRC check fails in SoC, the SoC SW shall transition to safe state within...

If the CRC check fails in SoC, the SoC SW shall transition to safe state within FRTI (200ms)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run system normally - Verify no CRC failures are detected - Inject CRC Error - Measure the time from CRC injection to PHY inactivity. - Verify this time is <=200 ms.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111289 - If the CRC check fails in SoC, the SoC SW shall transition to safe state within... is verified by : MVC-114973 - FUSA-PMIC-SOCCRCSAFE — SoC Safe-State Transition on Internal CRC Check Failure

MVC-99249 - After each write command to PMIC, the SoC SW shall read back the written value f...

After each write command to PMIC, the SoC SW shall read back the written value from the PMIC register and check this value against the commanded value- The process of writing, reading back and checking shall take less than 50ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system normally - Verify that all read-backs match the written values - Inject Fault for a PMIC Register - Resume execution. - Verify that the software performs the read-back and detect the mismatch. - verify time constraint
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95936 - The system shall ensure fault free 'fault free' reception of 'VideoStreamConfigu... duplicates : ~~MVC-100107~~ - Check configuration of power supply subsystem has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111288 - After each write command to PMIC, the SoC SW shall read back the written value f... is verified by : MVC-114051 - FUSA-PMIC-REGWRITECHECK Verification for PMIC Register (Normal Case)

MVC-99252 - If the SoC SW detects that an incorrect register value has been read back from P...

If the SoC SW detects that an incorrect register value has been read back from PMIC, the SoC SW shall re-write the register value within 50ms.

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system normally - write given value - halt execution of SoC SW task resposible for communication with PMIC - Inject Fault for a PMIC Register into the SoC - Resume execution. - Verify that the software performs the read-back (within time limits) and detect the mismatch and corrected.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97958 - The system shall ensure 'fault free' reception of 'FuSa-DiagnosticsInformation'... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111283 - If the SoC SW detects that an incorrect register value has been read back from P… is verified by : MVC-114052 - FUSA-PMIC-REGREWRITE — SoC Re-Write on First PMIC Register Read-Back Mismatch

MVC-99251 - If the SoC SW detects that an incorrect PMIC register value has been read back f...

If the SoC SW detects that an incorrect PMIC register value has been read back for the second time, the SoC SW shall transition to safe state within FRTI (200ms)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Run the system normally - write given value - halt execution of SoC SW task resposible for communication with PMIC - Inject Fault for a PMIC Register into the SoC - Resume execution. - Verify that the software performs the read-back and detects the mismatch. - Verify if the software performs the read-back - halt execution of SoC SW task resposible for communication with PMIC - Inject Fault for a PMIC Register into the SoC - Resume execution. - verify that system deactivates PHY - Measure the time from second detection to PHY become inactive is <= 200ms
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111282 - If the SoC SW detects that an incorrect PMIC register value has been read back f... is verified by : MVC-114053 - FUSA-PMIC-REGSAFESTATE — SoC Safe-State Transition on Second PMIC Register Read-Back Mismatch

MVC-99246 - The PMIC shall perform an analog built-in self-test (ABIST) during startup

The PMIC shall perform an analog built-in self-test (ABIST) during startup

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- power-up the system - verify that system is in 'normal' state - verify all ABIST (status)relevant register-bits are set to zero
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101110 - ABIST check is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114959 - FUSA-PMIC-ABISTSTARTUP — PMIC ABIST Executed and Passes During Startup

MVC-99245 - If ABIST fails, PMIC shall:

If ABIST fails, PMIC shall:

- Enter deep safe state during power-up (output rails shall be low), and re-run ABIST until ABIST passes (deep safe state via configuration of register PG_MAPPING 0xC6 bit 0)
- Pull ALERT# pin to low (configured via register ALERT_MAPPING 0xC4 bit 0)
- Pull PG to low (by default)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- inject ABIST fautl, e.g. exceed VOUT over-voltage (OV) for more than threshold duration (about 10µs), - Verify Output rails remain low (deep safe state). - verify that VOUT_OV_FAULT bit of the register is set to 0 - verify that the ABIST_FAULT bit of STATUS_MFR_SPECIFIC register is set to 1 - Verify ALERT# should go low and stay low. - Verify PG should go low and stay low.
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101109 - ABIST check unsuccessful is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-99248 - The PMIC shall perform a logic built-in self-test (LBIST) during startup

The PMIC shall perform a logic built-in self-test (LBIST) during startup

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	for sunny-day-scenario the following applies: - start the system - let system boot up - verfiy that PG is high - verify that STATUS_MFR_SPECIFIC[LBIST_FAULT] == 0
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101112 - LBIST check is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-99247 - If LBIST fails, PMIC shall:

If LBIST fails, PMIC shall:

- Have output rails stuck at low. POR shall not proceed.

- Pull PG to low

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	review of the PMIC documentation
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101111 - LBIST check unsuccessful is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-99315 - PMIC buck 3 and buck 4 outputs (Vout3 and Vout4) shall be 0.75V

PMIC buck 3 and buck 4 outputs (Vout3 and Vout4) shall be 0.75V

Note: Used for SoC VDDI

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start the system - wait for the boot up to complete - verify voltage value equals to 0.75V (+4.5% / - 3.5%)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... relates to : MVC-112498 - PMIC Bucks routing/naming changed - update requirements on System Level has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-99316 - The SoC SW shall configure PMIC at startup. PMIC buck 1 and buck 2 outputs shall...

The SoC SW shall configure PMIC at startup. PMIC buck 1 and buck 2 outputs shall be monitored for OV with +4.5% threshold and UV with -3.5% threshold using the following register configurations:

PG_CONFIG (0xCA) of Buck 1, bit[0] = 1 (Buck 1 thresholds: +4.5% / - 3.5%)

PG_CONFIG (0xCA) of Buck 2, bit[0] = 1 (Buck 2 thresholds: +4.5% / - 3.5%)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Normal Operation at nominal voltage, verify that both of PG pin and ALERT# pin are high - Decrease the output voltage of buck 1,2 (UV) - Verify that ALERT# pin and PG pin become low - Increase the output voltage of buck 1,2 (OV) - Verify that ALERT# pin and PG pin become low
Verification Method	Testable, checked by Test

Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... relates to : MVC-112674 - internal CR: additional voltage supervision on PMIC outputs has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111550 - The SoC SW shall configure PMIC at startup. PMIC buck 1 and buck 2 outputs shall... is verified by : MVC-113848 - FUSA-PMIC-OVUV1 PMIC Buck 1 OV/UV Protection (ALERT#/PG) Verified is verified by : MVC-113849 - FUSA-PMIC-OVUV2 — PMIC Buck 2 OV/UV Protection (ALERT#/PG) Verified

MVC-99320 - PMIC buck 1 output (Vout1) shall be 1.1V

PMIC buck 1 output (Vout1) shall be 1.1V
Note: Used for:
- DDR VDD2, VDDq
- PHY VDD_Core
- SoC MIPI_AVDD18_IO
- SoC DDR_DVDD_VDDQ
- SoC DDR_DVDD_IO

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start up the system - wait until starup completed - mesure voltage on VOUT1 pin of PMIC - verify that the voltage equals 1.1V within tolarance (ref. to MVC-99316)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... relates to : MVC-112498 - PMIC Bucks routing/naming changed - update requirements on System Level has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114534 - FUSA-PMIC-VOUT3 — PMIC Buck 3 Output Voltage Verified

MVC-99322 - The SoC SW shall configure PMIC at startup. PMIC buck 3 output shall be monitore...

The SoC SW shall configure PMIC at startup. PMIC buck 3 output shall be monitored for OV with +4.5% threshold and UV with -3.5% threshold using the following register configurations:
PG_CONFIG (0xCA) of Buck 3, bit[0] = 1 (Buck 3 thresholds: +4.5% / - 3.5%).

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Normal Operation at nominal voltage, verify that both of PG pin and ALERT# pin are high - Decrease the output voltage of buck 3 (UV) - Verify that ALERT# pin and PG pin become low - Increase the output voltage of buck 3 (OV) - Verify that ALERT# pin and PG pin become low
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... relates to : MVC-112674 - internal CR: additional voltage supervision on PMIC outputs has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111551 - The SoC SW shall configure PMIC at startup. PMIC buck 3 output shall be monitore... is verified by : MVC-113850 - FUSA-PMIC-OVUV3 PMIC Buck 3 OV/UV Protection (ALERT#/PG) Verified

MVC-99321 - PMIC buck 6 output (Vout6) shall be 0.8V

PMIC buck 6 output (Vout6) shall be 0.8V
Note: Used for SoC AVDD

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start up the system - wait until starup completed - mesure voltage on VOUT6 pin of PMIC - verify that the voltage equals 0.8V within tolarance (ref. to MVC-99317)

Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... relates to : MVC-112498 - PMIC Bucks routing/naming changed - update requirements on System Level has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114535 - FUSA-PMIC-VOUT4 — PMIC Buck 4 Output Voltage Verified

MVC-99324 - The SoC SW shall configure PMIC at startup. PMIC buck 4 output shall be monitore...

The SoC SW shall configure PMIC at startup. PMIC buck 4 output shall be monitored for OV with +4.5% threshold and UV with -3.5% threshold using the following register configurations:
PG_CONFIG (0xCA) of Buck 4, bit[0] = 1 (Buck 4 thresholds: +4.5% / - 3.5%)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Normal Operation at nominal voltage, verify that both of PG pin and ALERT# pin are high - Decrease the output voltage of buck 4 (UV) - Verify that ALERT# pin and PG pin become low - Increase the output voltage of buck 4 (OV) - Verify that ALERT# pin and PG pin become low
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... relates to : MVC-112674 - internal CR: additional voltage supervision on PMIC outputs has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111552 - The SoC SW shall configure PMIC at startup. PMIC buck 4 output shall be monitore... is verified by : MVC-113851 - FUSA-PMIC-OVUV4 PMIC Buck 4 OV/UV Protection (ALERT#/PG) Verified

MVC-99323 - PMIC buck 5 output (Vout5) shall be 1.8V

PMIC buck 5 output (Vout5) shall be 1.8V

Note: Used for:

- DDR VDD1
- PHY VDDIO
- Deserializer MAX96714 VDD18
- Deserializer MAX96714 VDDIO
- Flash VCC18

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start up the system - wait until starup completed - mesure voltage on VOUT5 pin of PMIC - verify that the voltage equals 1.8V within tolarance (ref. to MVC-99326)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114550 - FUSA-PMIC-VOUT5 — PMIC Buck 5 Output Voltage Verified

MVC-99326 - The SoC SW shall configure PMIC at startup. PMIC buck 5 output shall be monitore...

The SoC SW shall configure PMIC at startup. PMIC buck 5 output shall be monitored for OV with +4.5% threshold and UV with -3.5% threshold using the following register configurations:
PG_CONFIG (0xCA) of Buck 5, bit[0] = 1 (Buck 5 thresholds: +4.5% / - 3.5%)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI

Verification Criteria	- Normal Operation at nominal voltage, verify that both of PG pin and ALERT# pin are high - Decrease the output voltage of buck 5 (UV) - Verify that ALERT# pin and PG pin become low - Increase the output voltage of buck 5 (OV) - Verify that ALERT# pin and PG pin become low
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... relates to : MVC-112674 - internal CR: additional voltage supervision on PMIC outputs has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111553 - The SoC SW shall configure PMIC at startup. PMIC buck 5 output shall be monitore... is verified by : MVC-113854 - FUSA-PMIC-OVUV5 PMIC Buck 5 OV/UV Protection (ALERT#/PG) Verified

MVC-99325 - PMIC buck 2 output (Vout2) shall be 1.2V

PMIC buck 2 output (Vout2) shall be 1.2V

Note: Used for:

- Deserializer MAX96714 VDD
- Deserializer MAX96714 VTERM

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- start up the system - wait until starup completed - mesure voltage on VOUT2 pin of PMIC - verify that the voltage equals 1.2V within tolarance (ref. to MVC-99316)
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... relates to : MVC-112498 - PMIC Bucks routing/naming changed - update requirements on System Level has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is verified by : MVC-114591 - FUSA-PMIC-VOUT6 — PMIC Buck 6 Output Voltage Verified

MVC-99317 - The SoC SW shall configure PMIC at startup. PMIC buck 6 output shall be monitore...

The SoC SW shall configure PMIC at startup. PMIC buck 6 output shall be monitored for OV with +4.5% threshold and UV with -3.5% threshold using the following register configurations:

PG_CONFIG (0xCA) of Buck 6, bit[0] = 1 (Buck 6 thresholds: +4.5% / - 3.5%)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	- Normal Operation at nominal voltage, verify that both of PG pin and ALERT# pin are high - Decrease the output voltage of buck 6 (UV) - Verify that ALERT# pin and PG pin become low - Increase the output voltage of buck 6 (OV) - Verify that ALERT# pin and PG pin become low
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... relates to : MVC-112674 - internal CR: additional voltage supervision on PMIC outputs has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements is satisfied by : MVC-111554 - The SoC SW shall configure PMIC at startup. PMIC buck 6 output shall be monitore...

MVC-99319 - In case of OV or UV of any buck output, PMIC shall pull PG to low: Detection + r...

In case of OV or UV of any buck output, PMIC shall pull PG to low: Detection + reaction shall take no more than 50us.

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW

Verification Criteria	- Normal Operation at nominal voltage, verify that both of PG pin and ALERT# pin are high - Decrease the output voltage of any buck (UV) - Verify that PG pin become low - Increase the output voltage of any buck (OV) - Verify that PG pin become low
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... has parent : MVC-99078 - PMIC Safety Requirements is satisfied by : MVC-101113 - In case of OV or UV of any buck output, PMIC shall pull PG to low. is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-108190 - if a FuSa related error occurs in the PMIC, the PMIC shall pull ALERT# low. Detection + reaction time shall not take longer than 10ms

if a FuSa related error occurs in the PMIC, the PMIC shall pull ALERT# low. Detection + reaction time shall not take longer than 10ms

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	consult the PMIC documentation. Note: for the list of relevant FuSa errors refer to: - §5.3 Internal Safety Mechanisms of the "MPQ70165FS Safety Manual" Rev. 0.3 - FMEA of SOW2 for list on the System level
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-108191 - if a FuSa related error occurs in the PMIC, the PMIC shall pull PG to low. Detection + reaction time shall not take longer than the time that would cause the damage to any of the components of the SOW2.

if a FuSa related error occurs in the PMIC, the PMIC shall pull PG to low. Detection + reaction time shall not take longer than the time that would cause the damage to any of the components of the SOW2.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- startup the system - start timer - bring Buck Output Voltage to 0V - observe PG set to low - end timer - verfify elapsed time to be < 50 µs (ref. to §5.3.9 PG Under-Voltage (UV) Monitoring of the "MPQ70165FS Safety Manual" Rev. 0.3) Note: for the list of relevant FuSa errors refer to: - §5.3 Internal Safety Mechanisms of the "MPQ70165FS Safety Manual" Rev. 0.3 - FMEA of SOW2 for list on the System level
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-99318 - The OV and UV thresholds for all PMIC buck outputs shall be stored in flash

The OV and UV thresholds for all PMIC buck outputs shall be stored in flash

Status	Rejected
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	Checked by review
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97953 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97952 - The system shall check that the SG4 relevant 'VideoStreamConfiguration' was corr... satisfies : MVC-97957 - The system shall check that the 'VideoStreamConfiguration' was correctly written... has parent : MVC-99078 - PMIC Safety Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

2 Safe State Requirements

MVC-108192 - If the SoC SW detects ALERT# is set to low, the SoC SW shall transition into SafeState (detection: 50ms + reaction: 300ms)

If the SoC SW detects ALERT# is set to low, the SoC SW shall transition into SafeState (detection: 50ms + reaction: 300ms)

Status	Approved with assumption
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start up the system - start timer - set ALERT# to low - verify that system have entered a SafeState - stop timer - verify fullfilment of time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-100106 - Safe State Requirements is satisfied by : MVC-111285 - If the SoC SW detects ALERT# is set to low, the SoC SW shall transition into SafeState (detection: 50ms + reaction: 300ms) is related to : MVC-111539 - [TechSaReqs] Safe State Requirements is verified by : MVC-114974 - FUSA-PMIC-ALERTSOCSAFE — SoC SafeState Transition on ALERT# Low (Detection + Reaction Budget)

MVC-108193 - If PG is set to low, the SOW2 electronics shall transition the SOW2 into SafeState by cutting power off the PMIC...

If the SoC SW detects PG# is set to low, the SoC SW shall transition into SafeState (detection: 50ms + reaction: 300ms)

Note1: the main voltage monitoring element is a separate VMON chip.

Note2: PG set to low indicates some kind of FuSa relevant error of the PMIC. For the list of relevant FuSa errors refer to:

- §5.3 Internal Safety Mechanisms of the "MPQ70165FS Safety Manual" Rev. 0.3
- FMEA of SOW2 for list on the System level

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- start up the system - start timer - set PG to low - verify that system have entered a SafeState - stop timer - verify fullfilment of time constraints
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-100106 - Safe State Requirements is related to : MVC-105440 - [TechSaReqs] PMIC Safety Requirements

MVC-100138 - Entering the Safe State

If the system enters the Safe State, the system shall not transmit any image data via its Ethernet interface (the PHY link shall be disabled).

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run the system with video streaming. - Verify that Ethernet packets are being transmitted - Inject any fault to enter safe state - Verify that transmission stops within the expected FRTI
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100140 - SOW1 indicates a safety error to SOW2

If SOW1 indicates a safety error to SOW2, the system shall establish the Safe State within 150ms.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B

Allocation	SW
Verification Criteria	- Run the system with video streaming. - Force the safety error signal to its active state - Measure the time from the injection to the PHY becomes inactive - Verify this time is <= 150m
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100139 - Condition to remain in safe state

If the Safe State is established, the system shall remain in the Safe State as long as the condition that triggered the transition persists.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	- Run the system with video streaming. - Force the safety error signal to its active state - Verify that PHY becomes inactive and remains inactive - Remove the safety error signal - Verify that PHY becomes active again and resume normal operation
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100141 - Condition to remain in safe state

If a QM mechanism is provided to exit the Safe State, the mechanism shall ensure that any active Safe State condition overrides and prevents such exit.

Status	Predefined
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	Checked by review
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100142 - Safe State realization in SW

The SoC SW shall establish the Safe State by stopping the image sending activity of the Ethernet PHY.

This shall be achieved through at least one of the following:

- The SoC commands the PHY directly to disable link (AKA SW-LinkDisable-SafeSate)
- The SoC sets the PHY into RESET state (AKA SW-PHY_RESET-SafeSate)
- The SoC disables PMIC (AKA SW-PMIC_disable-SafeState)

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW

Verification Criteria	- Run the system with video streaming. - Verify that Ethernet packets are being transmitted - Inject fault provoking the SW-LinkDisable-SafeSate - Verify that is achieved by checking the PHY link being disabled and no link activity (interface SOW2-Vehicle) is present - Run the system with video streaming. - Verify that Ethernet packets are being transmitted - Inject fault provoking the SW-PHY_RESET-SafeSate - Verify that is achieved by checking the RST signal being asserted from SoC and no link activity (interface SOW2-Vehicle) is present - Run the system with video streaming. - Verify that Ethernet packets are being transmitted - Inject fault provoking the SW-PMIC_disable-SafeState - Verify that is achieved by checking the PMIC_EN signal being deasserted from SoC and no link activity (interface SOW2-Vehicle) is present
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-115145 - Safe State realization in HW

The system HW shall establish the Safe State by stopping the image sending activity of the Ethernet PHY.

This shall be achieved through at least one of the following:

- independent HW element that monitors the Voltage levels cuts off the power supply to the PMIC (AKA PWR_MAIN_disable-SafeState)
- independent temperature sensor disables PMIC OR the SoC HW disables PMIC (AKA HW-PMIC_disable-SafeState)

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Run the system with video streaming. - Verify that Ethernet packets are being transmitted - Inject fault provoking the PWR_MAIN_disable-SafeState - Verify that is achieved by checking the PWR_MAIN_EN signal being asserted by Voltage monitor and no link activity (interface SOW2-Vehicle) is present - Run the system with video streaming. - Verify that Ethernet packets are being transmitted - Inject fault provoking the HW-PMIC_disable-SafeState (high temperature) - Verify that is achieved by checking the PMIC_EN signal being asserted by Temperature sensor and no link activity (interface SOW2-Vehicle) is present - Run the system with video streaming. - Verify that Ethernet packets are being transmitted - Inject fault provoking the HW-PMIC_disable-SafeState (high temperature) - Verify that is achieved by checking the PMIC_EN signal being asserted by SoC HW and no link activity (interface SOW2-Vehicle) is present
Verification Method	Testable, checked by Test
Linked Work Items	has parent : MVC-100106 - Safe State Requirements

MVC-100144 - PMIC reaction on violation of its voltage monitoring thresholds

If the second PMIC detects a violation of its voltage monitoring thresholds, it shall establish the Safe State by performing system reset within 50ms.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Run the system with video streaming. - Increase or decrease the voltage against the nominal voltage value - Measure the time from the injection to the SoC becomes inactive - Verify this time is <= 50m
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100143 - Enable the second PMIC's watchdog by the Safety Core (Cortex R52)

Before the system begins transmitting any image data via the PHY, the Safety Core (Cortex R52) shall configure and enable the second PMIC's watchdog in Question Answer Mode (QA mode), with a timeout value appropriate for the system's fault detection intervals.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	Checked by review
Verification Method	Not testable, checked by Review
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100146 - Cases to enter the safe state by reset the SoC within 50 ms

After the watchdog is configured, the second PMIC shall reset the SoC within 50ms if any of the following occurs:

- The SoC responds with the wrong answer or
- The SoC responds later than 200ms after the last correct response.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HW
Verification Criteria	- Verify Watchdog Configuration (maximum response time is set to 200 ms) - Inject wrong answer - Measure the time from fault injection to SoC reset - Verify this time is <= 50m
Verification Method	Testable, checked by Test
Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100145 - When any safety relevant sub component of the SoC reports a safety error to the Safety Core (Cortex R52), the Safety Core shall establish the Safe State by stopping the PHY's image transmission

When any safety relevant sub component of the SoC reports a safety error to the Safety Core (Cortex R52), the Safety Core shall establish the Safe State by disabling the PHY's link (via direct command or by triggering the PMIC to reset the system) within 50ms.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	
Verification Method	
Linked Work Items	satisfies : MVC-97653 - When system cannot ensure that for each input video... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100147 - The Safety Core (Cortex R52) shall periodically monitor the error status of OSR components

The safety core (Cortex R52) shall periodically monitor the error status of other safety related (OSR) components:

- PMIC PG and ALERT signals
- PHY error indication
- Deserializer error indication (ERRB pin, Lock pin)

If any of these signals indicate a fault, the safety core shall treat it as a safety error and initiate the safe state (Triggering PMIC to reset the system).

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	
Verification Method	

Linked Work Items	satisfies : MVC-97954 - When the system cannot ensure data integrity of the SG4 relevant 'VideoStreamCon... satisfies : MVC-97951 - The system shall not transmit deteriorated stream for longer than 300ms by achie... satisfies : MVC-97946 - When the system cannot ensure that at the system output each video image is repe... satisfies : MVC-97653 - When system cannot ensure that for each input video... has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

MVC-100224 - To recover from CAMERA HEAD safe state the HOST SYSTEM shall switch OFF then ON the external power supply over the coaxial cable.

To recover from CAMERA HEAD safe state the SoC SW shall switch OFF then ON
the external power supply over the coaxial cable.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	HSI
Verification Criteria	
Verification Method	
Linked Work Items	satisfies : MVC-97947 - The system shall ensure 'fault free' reception of 'RAWVideoStream' from SOW1(cam... satisfies : MVC-95935 - The system shall ensure fault free 'fault free' reception of RAWVideoStream from... satisfies : MVC-95938 - The system shall ensure 'fault free' reception of RAWVideoStream from SOW1(camer... satisfies : MVC-95934 - The system shall ensure fault free 'fault free' configuration of RAWVideoStream... satisfies : MVC-95953 - The system shall ensure 'fault free' configuration of RAWVideoStream in SOW1. has parent : MVC-100106 - Safe State Requirements is related to : MVC-111539 - [TechSaReqs] Safe State Requirements

3 Production, Operation, Service and Decommissioning Requirements

MVC-100692 - All the safety-related configuration and calibration parameters for HW and SW co...

All the safety-related configuration and calibration parameters for HW and SW components shall be stored in flash sections protected with checksums.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	ASIL B
Allocation	SW
Verification Criteria	Review NVM map and verify that safety parameters are in protected sections.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-100616 - Production, Operation, Service and Decommissioning Requirements is related to : MVC-111540 - [TechSaReqs] Production, Operating, Service and Decommissioning Requirements

4 Requirements Towards Vehicle

MVC-100095 - After setting up the parameter

After setting up the parameter, the vehicle shall read back the parameter value and compare it with the intended value of this parameter (within application specific time (so that safety goals will not be violated, refer to Vehicle Safety Concept)).

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	
Allocation	
Verification Criteria	
Verification Method	
Linked Work Items	has parent : MVC-100094 - Requirements Towards Vehicle is related to : MVC-111541 - [TechSaReqs] Requirements towards vehicle

MVC-100111 - CV ECU shall be able to tolerate transient faults in VideoStream that occur within 100ms time span.

CV ECU shall be able to tolerate transient faults in VideoStream that occur within 100ms time span.

Status	Draft
Assignee(s)	Rekas, Miroslaw
ASIL	
Allocation	
Verification Criteria	
Verification Method	

Linked Work Items	has parent : MVC-100094 - Requirements Towards Vehicle is related to : MVC-111541 - [TechSaReqs] Requirements towards vehicle

ATTENTION: in case of image repetition (Frozen image) there is a built debouncing mechanism (100ms). This mechnism can lead to excessive latency in the system. BUT The latency check is perfomed before the image repetition takes place. This situation shall be handled on the Vehicle level based on time-stamp embedded into the video stream.

MVC-113533 - The vehicle shall put the SOW2 system through a complete power up/power down cycle

The vehicle shall put the SOW2 system through a complete power up/power down cycle (A full power up/power down cycle including VDDA_AO, asserting RST_N or wake-up from DEVICE_DEEP_SLEEP) within the latent fault tolerant time interval (L-FTTI), which is assumed to be 24 hrs.

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	SW
Verification Criteria	check the OEM documentation
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-100094 - Requirements Towards Vehicle

MVC-113534 - The Vehicle is able to handle frame loss or to implement appropriate safety measures against frame loss when required by the application. (e.g. readback, timestamping, message counter)

The Vehicle is able to handle frame loss or to implement appropriate safety measures against frame loss when required by the application. (e.g. readback, timestamping, message counter)

Status	Draft
Assignee(s)	
ASIL	ASIL B
Allocation	SW
Verification Criteria	check OEM documentation.
Verification Method	Not testable, checked by Review
Linked Work Items	has parent : MVC-100094 - Requirements Towards Vehicle

5 Requirements Towards SOW1

6 Temperature Sensing

7 FuSa AE/AWB Requirements

8 FuSa SOW1 configuration/requirements on SOW2

9 FuSa Vehicle requirements on SOW2

10 Recovered from Bin during Review

Recycle bin is no valid place for requirements because those requirements will still count towards statistics. Please reject invalid / not needed requirements with valid reason for traceability purposes.

11 SMs - feedback from FMEDA